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Joint Design for a Portable Coordinate Measuring Arm (PCMA)

Mechatronic Project 478
Final Report

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Executive summary

Title of Project
Joint Design for a Portable Coordinate Measuring Arm (PCMA)
Objectives
To derive technical performance measures (TPMs) from an error-propagation model, and use them to design, build, and test a position-holding joint that produces a constant, reliable holding torque.
What is current practice and what are its limitations?
Current joints rely on counterbalancing and lack holding torque, meaning the operator must physically support the arm's weight during operation.
What is new in this project?
The introduction of a position-holding mechanism specifically engineered using mathematically derived TPMs.
If the project is successful, how will it make a difference?
It will directly maintain or improve the physical measurement accuracy of the arm. Additionally, it will yield a modified kinematic model to serve as a generalized framework for future accuracy analysis.
What are the risks to the project being a success? Why is it expected to be successful?
Risks: University workshop manufacturing capabilities may fall short of required tolerances, and the holding mechanism could introduce unwanted friction. Expectation of Success: The design requirements are heavily grounded in an established, proven error model.
What contributions have/will other students made/make?
The project builds directly upon an initial error-propagation model developed by Stellenbosch alumna Leon Hall.
Which aspects of the project will carry on after completion and why?
The modified error-propagation model. It will be able to accept custom arm dimensions and compute the uncertainty distribution within a given work volume, forming the mathematical foundation for future PCMA development.
What arrangements have been/will be made to expedite continuation?

Extensive, standardized documentation will be left behind to ensure future researchers can build directly onto this work without repeating groundwork.

ECSA Exit Level Outcomes

This section addresses the following Engineering Council of South Africa (ECSA) exit level outcomes and indicates how each outcome was demonstrated in the project.

GA 1 - Problem-Solving

The project addresses an engineering problem which constitutes the lack of a joint design for Portable Coordinate Measurement Arms (PCMAs) that provides a repeatabile, reliable position-holding capability. This was solved by designing, modelling, and validating an electromagnetic brake joint capable of producing a sufficient holding torque.

GA 2 - Application Scientific and Engineering Knowledge

To satisfy this outcome, fundamental principles across mathematics, science, and computing must be applied to create effective solutions for complex engineering problems. This was demonstrated by using mathematical formulae to determine the required magnetic flux density, force generation, and torque produced to ensure the design achieved the sufficient torque. Principles of mechanics and physical science were used to reason and conceptualise the friction produced, the load-bearing capabilities, and the locking mechanics necessary to achieve reliable position-holding.

GA 3 - Engineering Design

This outcome requires the design of creative solutions for complex engineering problems and the design of systems, components, or processes to meet an identified need. This was achieved by executing a satisfactory design process that included a requirements analysis, concept generation and evaluation between pnematic and electromagnetic locking, as well as mechanical friction methods.

The design of the chosen electromagnetic brake concept was refined and modelled in detail, and complemented by a dedicated switching circuit. The design was finalised following a techno-analysis covering workshop fabrication costs and component availability.

GA 5 - Use of Engineering Tools

This outcome requires the demonstration of competence in creating, selecting, and applying appropriate techniques, resources, and modern engineering or IT tools. Autodesk Inventor was used for 3D CAD configuration, component packaging, and modelling the assembly of the PCMA joint. LTSpice was used to simulate and optimise the switching ciruit driving the electromagnet. Experimental test apparatus and measurements instruments were used to validate physical holding torque, while accounting for simulation idealisations and sensor limits.

GA 6 - Professional and Technical Communication

This outcome requires demonstrating the competence to communicate effectively on complex engineering activities, both orally and in writing. Written communication was demonstrated through a structured report adhering to departmental standards, supported by 3D CAD mechanical models, electronic circuit schematics, and empirical performance graphs. Oral and visual communication was demonstrated through the progress presentation, final oral presentation, and poster presentation designed for a general technical audience.

GA 8 - Individual and Collaborative Teamwork

This outcome requires demonstrating competence to function effectively as an indiviual. Although execution encountered significant schedule delays, individual competence was demonstarted by actively managing these setbacks, reevaluating critical path dependencies, and strategically prioritising core project objectives. Work was systematically refocused onto primary deliverables, specifically the design, driving circuit, and experimental validation of the electromagnetic joint. This was done while maintaining communication with the supervisor and coordinating with departmental workshop staff and laboratory technicians to bring the project to completion.

GA 9 - Independent Learning Ability

This outcome requires engaging in independent learning through self-directed research skills. Independent literature reviews were conducted electromag-

netic actuator design, high-current switching circuits, and joint stiffness optimisation. These were topics that extended beyond standard coursework. This self-acquired knowledge was synthesised and directly applied to design, build, and evaluate the final joint prototype.

Acknowledgements

I would like to express my gratitude to my supervisor, Prof. K. Schreve, for his guidance, expert insight, and continuous support throughout this project. Special thanks is extended to the technical staff of the Department of Mechanical and Mechatronics Engineering, as well as the Department of Electrical and Electronics Engineering for their assistance with prototype fabrication and component sourcing. Lastly, I am grateful to my family and peers for their constant encouragement and support throughout my studies at Stellenbosch University.

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List of symbols

Abreviations

3D	Three-dimensional
AI	Artificial Intelligence
AWG	American wire gauge
Back-EMF	Back electromotive force
CAD	Computer-Aided Design
D-H	Denavit-Hartenberg
DIP	Design Independent Parameter
DOF	Degrees-of-freedom
EC	Evaluation criteria
ECSA	Engineering Council of South Africa
EEPROM	Electrically erasable programmable read-only memory
EMF	Electromotive force
FBD	Freebody diagram
FEA	Finite element analysis
FR	Functional requirement
GPIO	General pin input/output
GPS	Global positioning system
HMI	Human machine interface
ISR	Interrupt service routine
MMF	Magnetomotive force
MOSFET	Metal-oxide-semiconductor field-effect transistor
NSoI	Narrower system of interest
PCMA	Portable coordinate measuring arm
PPR	Pulse per revolution
PSD	Position sensitive detector
SCADA	Supervisory control and data acquisition
SPAT	Single-point articulation test
SWG	Standard wire gauge

RHR	Right-hand rule
TPM	Technical performance measure
WSoI	Wider system of interest

Chapter 1

Introduction

1.1 Background

A portable coordinate measuring arm (PCMA) is a handheld metrology instrument used to capture three-dimensional (3D) coordinates directly from a physical object, as shown in Figure 1.1 (1).



Figure 1.1: A portable coordinate measurement arm (1).

PCMAs offer adaptable measurement capabilities across various engineering applications, including dimensional analysis, computer-aided design (CAD) inspection, reverse engineering, etc. (3). However, a primary limitation of conventional PCMAs is the absence of a position-holding mechanism. This leaves the arm vulnerable to positional instability and structural deformation during measurement or upon accidental impact (**?**). In addition to this, the instability complicates calibration procedures, which strictly require the arm to remain as still as possible throughout data acquisition (4).

As a result, integrating the design of an effective position-holding mechanism into the arm's joint addresses these limitations while enhancing operational safety. Such a joint design improves the kinematic stability of the arm, eliminates positional drift, and reduces operator fatigue.

1.2 Aim and Objectives

The aim of this project is to implement a position-holding mechanism on a PCMA to eliminate positional drift, enhance kinematic stability, and reduce operator fatigue during measurement and calibration.
The objectives of the project are:

1. To generalise the conventional Denavit-Hartenberg(D-H) model derived by Professor Schreve and Mr. Hall, with the purpose of error modelling (5).

2. To design, build and test key components of the joint to assess the joint design's mechanical feasibility.

1.3 Scope

In the process of determining D-H parameters, user manuals and available schematics will be used for information extraction. For the purposes of the project, the narrower system of interest (NSoI) is identified as the PCMA joint and the wider system of interest (WSoI)[1], as the NSoI, a reliable power source, as well as the preceding and succeeding link connected to joint in a controlled testing environment, as seen in Figure 1.2 below.

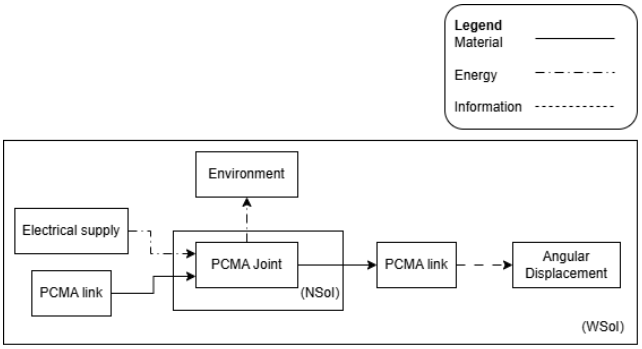


Figure 1.2: System Context

In addition to this, due to project timelines, physical implementation of the proposed design will be restricted to constructing critical subsystems necessary to validate the core mechanical, electrical, and software functionality.

[1]The NSoI is the system being designed and the WSoI includes the NSoI and the elements that the WSoI interacts with, (6)

1.4 Motivation

PCMAs are becoming more popular than traditional coordinate measuring machines (CMM) for in-line[2] and on-site industrial measurement (1). They have broad applicability across multiple industries such as aerospace, automotive, and heavy manufacturing, where having a portable measurement instrument decreases handling time, production delays, and revision of work.

Despite their growing popularity, the PCMA's revolute joints must satisfy competing demands, including:

- Tight run-out tolerances for spatial accuracy.
- Sufficient holding torque to keep the arm stationary when required.
- Internal cable routing.

This has proved difficult since no commercially available joint designs have been found to meet these demands.

Additionally, it has been established that angular position errors at each joint propagate through the kinematic chain, which affects end-effector's accuracy. This proves that joints have a measurable impact on the PCMA's accuracy (7). This project addresses the knowledge gap by deriving joint specifications from a generalised model based on an error model that was developed by a Stellenbosch Alumnus, Leon Hall (5). This will produce a validated joint and testing methodology to support future PCMA development.

1.5 Use of AI Tools

Gemini is an artificial intelligence (AI) tool that was used to check grammatical errors and sentence structure as well as LaTex formatting for the report.

1.6 Report Overview

This section follows the following structure:

- Chapter 1 introduces the project background, aim and objectives, scope, motivation for the project, and report structure.
- Chapter 2 presents a literature review discussing sources relevant to PCMA joint context, the Denavit-Hartenberg (D-H) kinematic model, the MATLAB error-propagation uncertainty model, and electromagnetic brake design theory.

[2] In the context of PCMAs, "in-line" refers to performing quality inspections and measurements directly on the production shop floor, instead of a temperature-controlled quality laboratory, (1).

- Chapter 3 details the stakeholder identification, design independent parameters, functional requirements, and technical performance measures (TPMs) for the PCMA joint.
- Chapter 4 discusses concept development and evaluation for the joint, in which an electromagnetic brake concept is selected and refined.
- Chapter 5 describes the design analysis of the chosen concept across its mechanical, electrical, and software design domains.
- Chapter 6 presents the as-built subsystem construction, safety protocols, experimental testing, holding force verification, and a discussion of key electromechanical constraints.
- Chapter 7 outlines the end-of-life protocol, covering design for disassembly, material recycling streams, hazardous substance mitigation, and circular economy strategies.
- Chapter 8 presents the main conclusions of the report.
- Appendices include the itemised bill of materials, detailed design calculations and LTspice simulations, engineering drawing pack, and project schedule.

Chapter 2

Literature review

2.1 PCMA Context

As earlier stated, PCMAs are precise, handheld metrology instruments used to capture 3D measurements directly from physical objects. These devices are built with components that operate similarly to an elbow, forearm, and wrist (7). Typically, an arm operates with six to eight degrees-of-freedom (DOF) to reach inaccessible areas. The measurement accuracy of a PCMA relies heavily on a high-precision probe as an end-effector and high-resolution angular encoders located at each rotating joint (7). When an operator touches an object through the probe, the arm's movement is tracked in real time by the angular encoders (1). These measurements are used calculate and plot the exact 3D coordinates of the object within the PCMA's reference coordinate system (1). More advanced PCMA models may include internal temperature sensors to compensate for environmental temperature changes. They may also incorporate laser-line scanners for non-contact, high-frequency data capture (7).

It is noted that PCMAs offer significant advantages in flexibility and portability, however, their measurement accuracy is dependent on a rigorous calibration procedure (8). This procedure is used to identify D-H parameters (which will be discussed further Section 2.2) and account for them to improve the accuracy of the PCMA, allowing the PCMA's internal software to compensate for error in end-effector measurement results (9). Without proper calibration, the inherent positioning error in the multi-joint arm would exacerbate the arm's measurement uncertainty. This would result in significantly less accuracy in PCMAs (9).

A few authors have highlighted several methodologies and challenges in PCMA calibration to address structural limitations. Alvarez et al. (9) explains that calibration utilises the D-H method to model geometric coordinate transformations between consecutive reference systems. They make use of non-linearisation algorithms to minimise measurement errors. Their research features novel data aqcusition methods that use laser-tracking multilateration to achieve accurate nominal coordinates across the arm's working volume.

Laser-tracking multilateration is a metrology technique that determines spatial position by measuring interferometric distances from multiple fixed tracking locations to a reflector on the end-effector (9). In simpler terms, exact straight-line distances can be determined by the splitting and recombination of the laser at fixed stations to a reflector on the probe (9). By computing where these distances intersections, the system determines where the probe's true 3D position in space is with micrometric accuracy (9). This creates a reference map to identify and correct errors in the PCMA's internal D-H kinematic model (9).

In addition to this, Alvarez et al. (9) note that human factors introduce non-uniform contact forces and variable operating trajectories due to their manual operation. Through finite element analysis (FEA) and multibody dynamics, they demonstrate that human-driven forces induce elastic, dynamic, structural deformations within the arm during probing. Due to the linearity of these deformations, Alvarez et al. (9) conclude that calibration models must account for applied pressure in order to improve overall accuracy. Ibaraki and Saito (3) note that overall measurement uncertainty of PCMAs is aggrevated by angular position errors within the rotary axes, which often stem from graduation errors[1] of the rotary encoder scale. To minimise this effect, Ibaraki and Saito (3) employ the single point articulation test (SPAT) and modify it with 3D displacement sensors to dynamically identify these angular errors.

Despite these advancements, the proposed solutions of the aforementioned authors, including Alvarez et al. (9) and Ibaraki and Saito (3), focus solely on calculating and compensating for the error computationally. There is insufficient exploration into mechanical stablisation techniques to address the positional error observed. Such an approach would prevent structural elastic deformations at the source, instead of relying on post-measurement software correction.

2.2 The Denavit-Hartenberg (D-H) Model

A Denavit-Hartenburg (D-H) model is a tranformation matrix that models an end-effector's position and orientation relative to the system of articulate linkages it is connected to (5). This is done by relating the end-effector's reference coordinate system to the last linkage's reference coordinate system through the matrix (5). Hall (5) conducted a full investigation of measurement error in PCMAs, which also accounted for spatial error using a modified D-H model. The motivation behind this investigation was that PCMA manufacturers did not adopt a common standard that properly governed their products' accuracy (5). It was found that there was a descrepancy betweem the advertised PCMA accuracy to the actual PCMA accuracy as manufacturers did not account for spatial

[1] Graduation inaccuracy refers to microscopic physical imperfections in the spacing, placement, or width of the measurement lines (graduations) inscribed on a rotary encoder's internal scale.

dependancy in their product (5).

The first step of the project was to derive a D-H model, based on the FaroArm QuantumX, which is a six DOFs PCMA. The relative coordinate systems for each joint along the length of the arm is defined using a Cartesian coordinate system (X, Y, Z), where rotations and translations follow the right-hand rule (RHR)[2] (5). In this system, the Z-axis is assigned directly along the joint's axis of rotation (5). The X-axis is drawn along the common normal which refers to the shortest, perpendicular line between the previous and current Z-axes (5).

The PCMA was modelled as a serial kinematic chain based on its architecture using this coordinate system. The end effector's position is seen in Equation 2.1:

$$T_6^0 = A_1 \cdot A_2 \cdot A_3 \cdot A_4 \cdot A_5 \cdot A_6 \tag{2.1}$$

wherein, each A_i models the position of each joint relative to the previous joint's reference coordinate system, as seen in Equation 2.2:

$$A_i = \begin{bmatrix} \cos\theta_i & -\sin\theta_i \cos\alpha_i & \sin\theta_i \sin\alpha_i & a_i \cos\theta_i \\ \sin\theta_i & \cos\theta_i \cos\alpha_i & -\cos\theta_i \sin\alpha_i & a_i \sin\theta_i \\ 0 & \sin\alpha_i & \cos\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2.2}$$

where (5):

- a_i is the joint length, which is the shortest distance between the Z_{i-1} axis and the Z_i axis measured along the common normal axis (X_i).
- θ_i is the rotational angle, which the angle from the X_{i-1} axis to the X_i axis, measured about the Z_{i-1} axis using the RHR.
- α_i is the twist angle, which is the angle from the Z_i axis to the Z_i axis, measured about the common normal axis (X_i) using the RHR.
- d_i the joint offset, which is the distance from the origin of joint $i-1$ to the origin of joint i, measured along the Z_{i-1} axis.

Figure 2.1 below illustrates these parameters and their geometric relationships on two (for legibility) of the joints on the FaroArm QuantumX.

[2]The RHR dictates that if you align your right thumb (representing the positive Z_i-axis) with the positive direction of rotation, your index finger will point along the positive X-axis and your middle finger will bend toward the positive Y-axis.

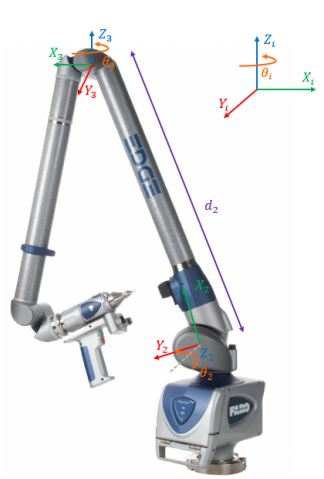


Figure 2.1: D-H Parameters Modelling a Pitching (Joint 2) and Rolling Joint (Joint 3)

It is important to note that the twist angle, α_i, is omitted from the visual layout to maintain legibility. This is because the relative orientation between Z_{i-1} and Z_i is fixed according to the principals of the model (5). Together, in Equation 2.2 shown above, they define how a link translates and rotates from one joint's frame of reference to the next.

The D-H parameters can now be determined for the PCMA. The nominal D-H parameters of the arm are not published as they are stored in the device's internal electrically erasable programmable read-only memory (EEPROM) firmware or stored within its proprietary device drivers (10). However, they can be derived from the physical dimensions in the product specifications document due to the analogous structure of the FaroArm QuantumX family (10). The specific length chosen was the three-meter arm and its D-H parameters are tabulated under Table 2.1 below (10):

Table 2.1: Denavit-Hartenberg Parameters for the FaroArm Model

Link i	a_i **(mm)**	α_i	d_i **(mm)**	d_i **Notes**
1	0	+90°	283	Offset as a result of the base
2	0	-90°	755	Length of link 1 from position-sensitive detector
3	0	+90°	0	Intersecting elbow axes[3]
4	0	-90°	569	Length of link 2 from PSD
5	0	+90°	0	Intersecting wrist axes[4]
6	0	-90°	220	End-effector length

[1]Physically, this represents the twist and hinge encoders of the elbow working together as a multi-axis pivot, instead of being separated by a distinct tubular segment.

[2]There is a physical housing that connects the encoders, which reduces the link length to zero, creating a joint-link behavior before the end-effector offset.

As shown by Table 2.1, for the arm under analysis, $a_i = 0$ across all links. This is due to every consecutive pair of joints intersecting directly at a single joint or lying co-planar with the structural axes. As a result, there is no perpendicular offset or lever arm that separates the adjacent rotational centerlines. By contrast, there are non-zero values for some of the joint offsets (d_i) to represent where a physical extension occurs along the joint axis. The D-H model resulting from these values can be seen in Figure 2.2 below.

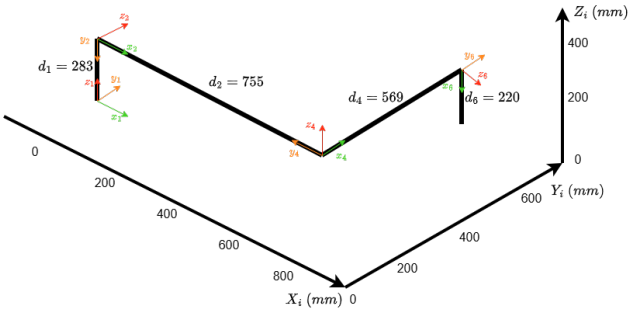


Figure 2.2: D-H Model of the FaroArm QuantumX Arm

By substituting the tabulated D-H parameters, the following joint positions can be determined by Equation 2.3

$$A_1 = \begin{bmatrix} \cos\theta_1 & 0 & \sin\theta_1 & 0 \\ \sin\theta_1 & 0 & -\cos\theta_1 & 0 \\ 0 & 1 & 0 & 283 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad A_2 = \begin{bmatrix} \cos\theta_2 & 0 & -\sin\theta_2 & 0 \\ \sin\theta_2 & 0 & \cos\theta_2 & 0 \\ 0 & -1 & 0 & 755 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_3 = \begin{bmatrix} \cos\theta_3 & 0 & \sin\theta_3 & 0 \\ \sin\theta_3 & 0 & -\cos\theta_3 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad A_4 = \begin{bmatrix} \cos\theta_4 & 0 & -\sin\theta_4 & 0 \\ \sin\theta_4 & 0 & \cos\theta_4 & 0 \\ 0 & -1 & 0 & 569 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A_5 = \begin{bmatrix} \cos\theta_5 & 0 & \sin\theta_5 & 0 \\ \sin\theta_5 & 0 & -\cos\theta_5 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad A_6 = \begin{bmatrix} \cos\theta_6 & 0 & -\sin\theta_6 & 0 \\ \sin\theta_6 & 0 & \cos\theta_6 & 0 \\ 0 & -1 & 0 & 220 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (2.3)$$

Note that under normal kinematic modelling assumptions, the parameters a_i, α_i, and d_i are treated as constants determined by fixed mechanical geometry. This would include tubing, piping, bearing housings, etc. Alternatively, the joint angle, θ_i, is the only kinematic variable because it is measured dynamically during operation via sensory equipment, such as rotary encoders. As a result, in an idealised kinematic model, encoder resolution, noise, and quantisation errors

in θ_i represent a source of uncertainty, which propagates to the end-effector, through the kinematic chain.

The A_i matrices are input into the error propagation model to determine the position uncertainty distribution of the PCMA's end-effector across the workspace, using Equation 2.1. These calculations can be cumbersome if done by hand, however, MATLAB was chosen to compute them which will be discussed in the following section.

2.3 The Error Propagation Model

As mentioned in the previous section, an error propagation model was established to quantify the kinematic spatial accuracy of the PCMA and map its uncertainty across the entire working volume. This mapping enabled the derivation of optimal encoder performance measures through a requirements analysis conducted under Chapter 3.

As explained by Hall (5), this model serves as a mathematical framework depicting how known uncertainties move through an articulated kinematic chain and how these uncertainties affect the output positioning error of the end-effector (5). The output-position uncertainty is calculated using Equation 2.4:

$$\Lambda_F = J(x) \cdot \Lambda_x \cdot J(x)^T \tag{2.4}$$

where $J(x)$ is the Jacobian matrix of the end-effector's position (derived from the transformation matrix in Equation 2.1) and Λ_x is the input uncertainty covariance matrix (5).

Under nominal conditions, Λ_x is a comprehensive $(4n \times 4n)$ block-diagonal covariance matrix, where n is the number of DOF, intended to model the variance across all D-H parameters (including a_i, θ_i, α_i, and d_i), as shown in Equation 2.5 (5):

$$\Lambda_x = \text{diag}(\boldsymbol{\Sigma}_1, \boldsymbol{\Sigma}_2, \dots, \boldsymbol{\Sigma}_n) \quad \text{where} \quad \boldsymbol{\Sigma}_i = \begin{bmatrix} \sigma_{a_i}^2 & \sigma_{a_i,d_i} & \sigma_{a_i,\alpha_i} & \sigma_{a_i,\theta_i} \\ \sigma_{d_i,a_i} & \sigma_{d_i}^2 & \sigma_{d_i,\alpha_i} & \sigma_{d_i,\theta_i} \\ \sigma_{\alpha_i,a_i} & \sigma_{\alpha_i,d_i} & \sigma_{\alpha_i}^2 & \sigma_{\alpha_i,\theta_i} \\ \sigma_{\theta_i,a_i} & \sigma_{\theta_i,d_i} & \sigma_{\theta_i,\alpha_i} & \sigma_{\theta_i}^2 \end{bmatrix} \tag{2.5}$$

These parameter variances usually account for structural deformations, manufacturing tolerances, and environmentally-induced effects, such as ambient thermal expansion altering link lengths (a_i, d_i) or joint alignments (α_i) that diminish a PCMA's accuracy (5). However, as stated under Section 1.3, it is assumed that the PCMA operates within a controlled testing environment, thereby maintaining a_i, d_i, and α_i as constant deterministic values $(\sigma_{a_i}^2 = \sigma_{d_i}^2 = \sigma_{\alpha_i}^2 = 0)$ (5). As a result, the input covariance matrix simplifies to a reduced $(n \times n)$ joint angle covariance matrix, Λ_θ:

$$\Lambda_\theta = \text{diag}(\sigma_{\theta_1}^2, \sigma_{\theta_2}^2, \dots, \sigma_{\theta_n}^2) \tag{2.6}$$

By assuming independent and identically distributed joint encoder errors, Λ_θ scales with a baseline angular standard deviation, typically set to an industry-standard nominal value of $\sigma_\theta = 6$ arcsec.

The resulting end-effector covariance matrix Λ_F characterizes the multi-variate variance of the position vector $[X, Y, Z]^T$ (5). From Λ_F, the standard deviation along principal axes ($\sigma_F = \sqrt{\text{tr}(\Lambda_F)}$) is extracted. To express boundaries with high statistical confidence, the expanded spatial uncertainty at a 95% confidence level (U_{95}) is evaluated as:

$$U_{95} = k \cdot \sigma_F \tag{2.7}$$

where $k = 2$ corresponds to the coverage factor for a normal distribution.

2.3.1 MATLAB Uncertainty Mapping

A key constraint of Equation 2.4 is that the spatial positioning uncertainty is posture-dependent and yields an error value for only one discrete pose of the PCMA at a time. To evaluate the global kinematic performance of the FaroArm QuantumX, specifically the 3 m arm, a posture sampling scheme was implemented in MATLAB across the wokring volume.

The workspace mapping algorithm follows a structured numerical pipeline, starting with the kinematic parameterization and setup. As shown in Figure 2.3, the code begins by loading the nominal D-H parameters (a_i, d_i, α_i) of the 6-DOF FaroArm QuantumX.

```
%% 1. Define DH model

% FaroArm Quantum 21000, size 3000 mm
L1 = 755;   % mm
L2 = 569;   % mm

% Simplified DH geometry for the 3000 mm arm
% a_i are set to zero because perpendicular joint-axis offsets are not
% included in this simplified geometric approximation.
a = [0 0 0 0 0 0];

% d_i contains the visible base, arm, forearm, and probe dimensions
d = [283 L1 0 L2 0 220];

% Twist angles
% Recommended nominal option when following Hall & Schreve's AACMM DH pattern:
alpha_deg = [90 -90 90 -90 -90 180];

% If you want to test your earlier simplified assumed twist sequence, use:
% alpha_deg = [90 -90 90 -90 90 0];

alpha = deg2rad(alpha_deg);

% Number of DH links / joints
n = numel(a);

% Basic validation
if numel(d) ~= n || numel(alpha) ~= n
    error('DH parameter vectors a, d, and alpha must have the same length.');
end

fprintf('\n--- DH model definition ---\n');
fprintf('Number of DH links / joints: %d\n', n);

DH_table = table((1:n).', a(:), alpha_deg(:), d(:), ...
    'VariableNames', {'Link','a_mm','alpha_deg','d_mm'});

disp(DH_table);
```

```
--- DH model definition ---
Number of DH links / joints: 6
    Link    a_mm    alpha_deg    d_mm
    ____    ____    _________    ____

     1       0          90       283
     2       0         -90       755
     3       0          90         0
     4       0         -90       569
     5       0         -90         0
     6       0         180       220
```

Figure 2.3: Error-Propagation Variable Input

The next step is to initialise the covariance matrix. As aforementioned, by assuming independent angular measurement errors with equal variance across all joints, the joint covariance matrix Λ_θ can be defined using an identity matrix scaled by the encoder variance. As seen in Figure 2.4, an industry-standard nominal encoder accuracy of 6 arcsec ($\sigma_\theta = 6''$) is specified.

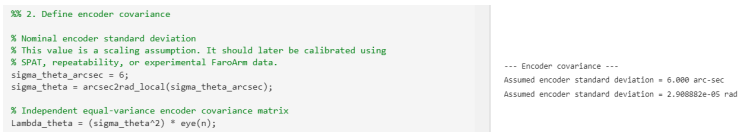


Figure 2.4: Error-Propagation Encoder Covariance Input

The next step is to discretise the working volume and evaluate the poses the volume. To sample the reachable envelope, allowable angular ranges ($\theta_1 \ldots \theta_6$) are discretized into uniform step increments, as shown in Figure 2.5.

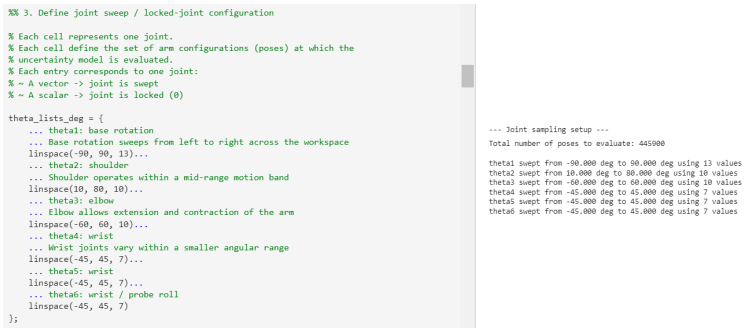
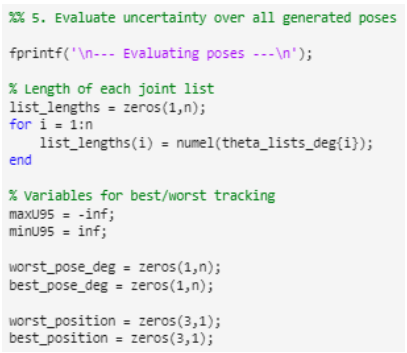
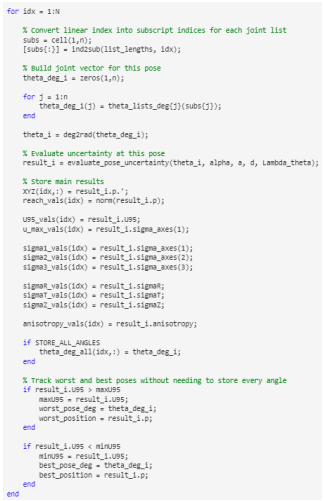


Figure 2.5: Joint Sweep Discetisation

These discrete joint-angle combinations generate a gridl'm uss of spatial postures spanning the entire working volume. For each discrete pose, the algorithm evaluates forward kinematics to compute 3D Cartesian end-effector coordinates and determines the analytical Jacobian matrix $J(x)$, as shown in Figure 2.6.



(a)



(b)

Figure 2.6: Numerical evaluation of kinematic poses and Jacobian transformation across the sampled workspace.

Next, Equation 2.4 is evaluated at every spatial node yields the end-effector covariance matrix Λ_F, from which the 95% expanded spatial uncertainty interval (U_{95}) is computed. The resulting spatial map is illustrated in Figure 2.7.

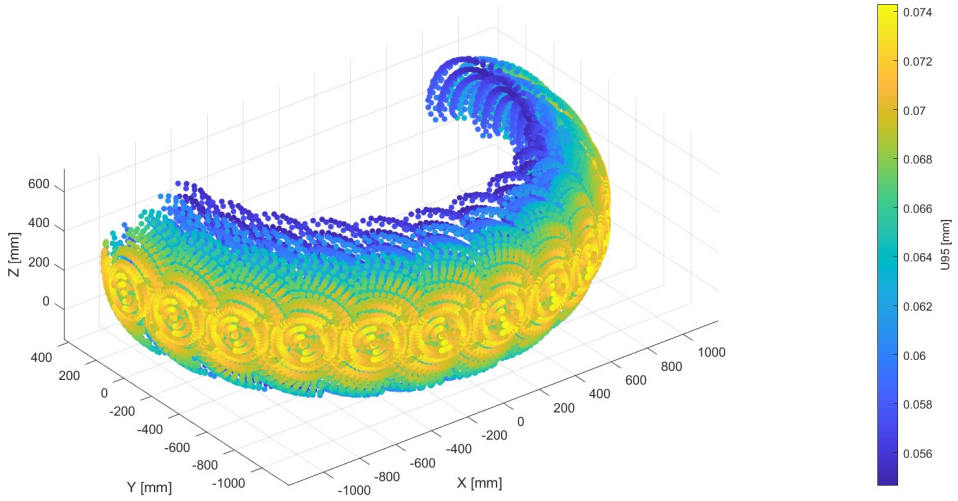


Figure 2.7: Spatial U_{95} uncertainty distribution across the FaroArm Working Volume

The spatial map demonstrates the posture-dependence of kinematic accuracy across the working envelope. Optimal performance occurs when the arm is folded near its base (highlighted in deep blue), achieving a minimum U_{95} spatial uncertainty of approximately 56 µm. Conversely, high kinematic leverage at full reach amplifies angular encoder noise, increasing U_{95} to a maximum of 74 µm at the outer boundary (highlighted in bright yellow).

Furthermore, the generalizability of this model enables joint-locking simulations. By setting targeted variance elements in Λ_θ to zero (simulating active position-holding), the framework can evaluate trade-offs for optimized multi-DOF arm configurations in future PCMA developments.

2.4 Electromagnetic Brake Design Theory

2.4.1 Brake Design Theory

The design of a manual position-holding joint mechanism for a PCMA integrates principles from brake design theory, electromagnetic field dynamics, physical actuator construction, and electronic switching circuits based on the chosen concept (as will be discussed in Chapter 4). In standard industrial machinery, brake design must accommodate heavy thermal loads generated by continuous friction, requiring large surface area ratios between the brake disc and friction plate to maximize heat dissipation (2). However, because the PCMA joint mechanism operates statically or quasi-statically, heat dissipation constraints

can be safely neglected. The joint brake is engaged mainly to hold a fixed position against gravity or support manual operation, instead of continuously decelerating a high-inertia motorized system. As a result, the area ratio between the brake disc and the friction plate can be made $1:1$ without thermal failure risks (2).

According to Shigley's Mechanical Engineering Design (2), under the uniform pressure assumption, the clamping force F applied by the electromagnetic actuator, seen in Figure 2.8, distributes a uniform pressure p across the annular friction interface, shown by Equation 2.8:

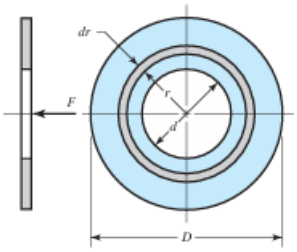


Figure 2.8: Disk Friction Member (2)

$$F = \frac{\pi p_a}{4} \cdot (D^2 - d^2) \tag{2.8}$$

where r_o is the outer radius, r_i is the inner radius, and F is the force applied by the actuation of the brake.

2.4.2 Electromagnetic Field Dynamics and Force Generation

Once the required axial holding force is determined, electromagnetic theory dictates how electrical power generates that force. Consider a solenoidal magnetic circuit comprising two aligned cylindrical ferromagnetic cores (pole pieces) of cross-sectional area A_c, separated by a narrow axial working air gap g and energized by a continuous helical coil with N total turns carrying current I, as seen in Figure 2.9 (11).

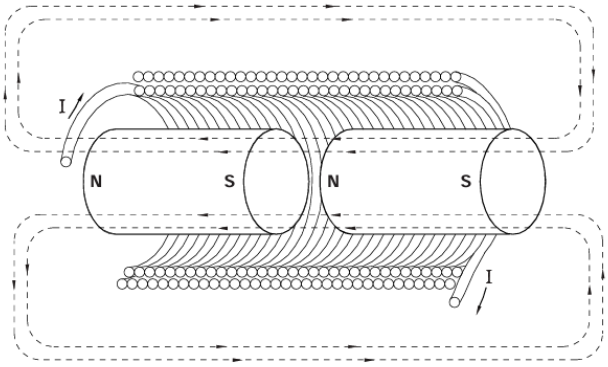


Figure 2.9: Solenoid

Applying Ampère's Law around the closed magnetic flux path (represented by the dashed lines looping through both cores) yields the total magnetomotive force (MMF), $\mathcal{F} = NI$ (11). Assuming high magnetic permeability for the ferromagnetic core ($\mu_r \gg 1$), the reluctance of the iron cores ($\mathcal{R}_{\text{core}} = \frac{l_{\text{core}}}{\mu_0 \mu_r A_c}$) is negligible compared to the reluctance of the non-magnetic working air gap ($\mathcal{R}_g = \frac{g}{\mu_0 A_c}$) (11). The total circuit reluctance is thus dominated by the gap, calculated by Equation 2.9 (11):

$$\mathcal{R}_{\text{total}} \approx \mathcal{R}_g = \frac{g}{\mu_0 A_c} \tag{2.9}$$

The total magnetic flux Φ traversing the axial air gap between the adjacent north (N) and south (S) pole faces is expressed via Hopkinson's Law ($\mathcal{F} = \Phi \mathcal{R}_{\text{total}}$), expressed by Equation 2.10 (11):

$$\Phi = \frac{NI}{\mathcal{R}_g} = \frac{\mu_0 A_c NI}{g} \tag{2.10}$$

As a result, the uniform magnetic flux density B established within the working air gap is defined by Equation 2.11 below (11):

$$B = \frac{\Phi}{A_c} = \frac{\mu_0 NI}{g} \tag{2.11}$$

where $\mu_0 = 4\pi \times 10^{-7}$ H/m is the permeability of free space (11).

The electromagnetic clamping force, F, pulling the core faces together across the central gap interface is derived from the field energy density $u_m = \frac{B^2}{2\mu_0}$ stored within the volume of the air gap $V_g = A_c g$ (11). Differentiating the field energy $W_m = u_m V_g = \frac{B^2 A_c g}{2\mu_0}$ with respect to gap displacement g yields the Maxwell stress tensor force formulation, seen in Equation 2.12 below (11):

$$F = k\frac{\partial W_m}{\partial g} = k\frac{B^2 A_c}{2\mu_0} = k\frac{\mu_0 A_c (N_{\text{solenoid}} I_{total})^2}{2g^2} \tag{2.12}$$

where k is the number of solenoids used in under A_c, N_{solenoid} is the number of coils for one solenoid, and I_{total} is the total current used to power all the solenoids.

Ferromagnetic core materials exhibit a non-linear B-H magnetization relationship governed by the alignment of internal magnetic domains in the metal, as illustrated in Figure 2.10 (11).

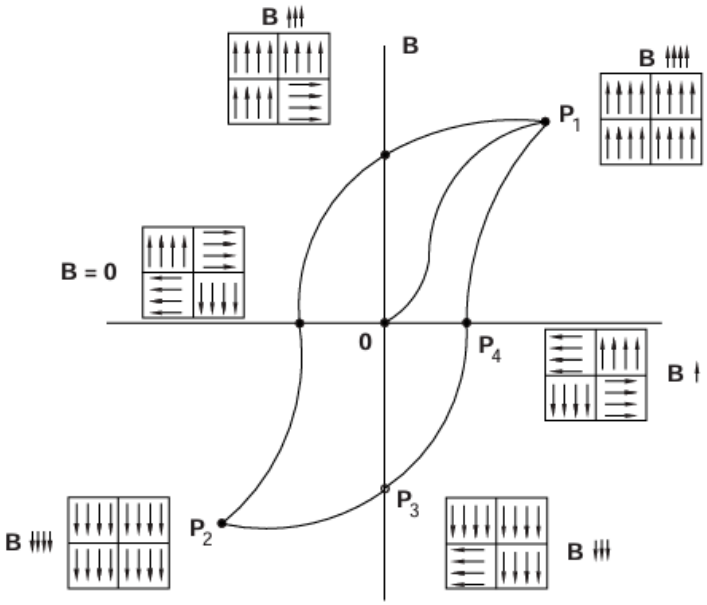


Figure 2.10: Progression of Magnetic Flux Saturation and Hysteresis Domain Alignment

Starting from an unmagnetized state at the origin (0), magnetic domains are randomly oriented, yielding zero net flux density ($B = 0$) (11). As the magnetomotive force increases, domains rotate and align with the field, tracing the initial magnetization curve up to point P_1. At P_1, magnetic saturation (B_{sat}) is achieved, where virtually all magnetic domains are fully aligned parallel to the applied field (11). Beyond P_1, further increases in coil excitation (NI) produce diminishing returns in flux density ($\Phi_{\text{max}} = B_{\text{sat}} A_c$), placing a theoretical upper limit on the achievable electromagnetic clamping force $F \propto B^2 A_c/(2\mu_0)$ (11).

When the excitation current is subsequently removed, the core does not return to origin 0 (11). Instead, residual domain alignment causes a non-zero residual magnetic flux density (B_r) to persist along the vertical axis. This would create an unwanted residual clamping force that prevents immediate release of the PCMA joint brake (11). Demagnetizing the core requires applying a reverse field corresponding to the coercive force at point P_4 (11). To ensure rapid mechanical disengagement upon de-energization, low-carbon soft magnetic steel with a narrow hysteresis loop, characterized by low coercive force ($P_4 \rightarrow 0$) and minimal remanence ($B_r \rightarrow 0$), must be selected for the core and yoke elements (11).

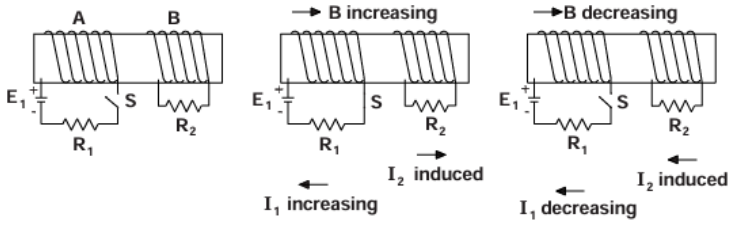


Figure 2.11: Depicting the Increase and Decrease of Magnetic Force

2.4.3 Electromagnet Actuator Construction and Parameter Balance

To bridging theoretical equations with physical construction, Chapters 4 and 5 of the HydraForce Solenoid Training Manual (11) outline the core parameter trade-offs required to maximize magneto-motive force (NI) within tight spatial constraints. Increasing the coil turn count N increases wire resistance R, which in turn reduces the current I drawn from a fixed voltage supply ($V = IR$), seen in Figure 2.11 below. In contrast, increasing wire diameter lowers resistance R to allow higher current I, but increases total coil volume and thermal dissipation requirements ($P = I^2R$). Furthermore, as current flows through the copper windings, the rising temperature increases electrical resistance according to Equation 2.13:

$$R_T = R_{20}\,[1 + 0.00393(T - 20°\text{C})] \tag{2.13}$$

where R_{20} is the coil resistance at 20°C and T is the operating temperature in degrees Celsius. Practical component selection requires the outer yoke or shell to be constructed from low-carbon steel due to its high magnetic permeability and minimal residual magnetism retention. Flat-face armatures are preferred over conical or proportional shapes because they maximize holding force at near-zero air gaps ($g \to 0$).

Chapter 3

Requirement Analysis

The objective of the requirement analysis is to establish clear expectations of the PCMA's joint design for the purposes of concept generation and development.

3.1 Identification of Stakeholders

The first step in the requirement analysis process is to identify stakeholders from the operational and support life cycle phases of the project. Stakeholders will be classified into the following categories, as defined below:

- Primary stakeholders directly interact with the system while it is in operation.
- Secondary stakeholders contribute to the development and functioning of the system.
- Indirect stakeholders rely on or are affected by the system, but do not interact with the system directly.

The stakeholders relevant to the project are defined in Table 3.1 below.

Table 3.1: Identified stakeholders and their classifications

Stakeholder ID	**Description**	**Classification**
SID_01	Supervisor of the project	Primary
SID_02	Students researching PCMA development	Secondary
SID_03	Companies invested in PCMA development	Indirect
SID_04	Mechanical workshop staff	Indirect
SID_05	Mechatronics laboratory managers	Indirect

3.2 Design Independent Parameters

The next step is to define the design independent parameters (DIPs). DIPs influence the system during the operational and support life cycle stages, but are not controllable nor are they part of the system being defined (6). The DIPs of the joint design are detailed in Table 3.2 below.

Table 3.2: Design Independent Parameters

DIP_ID	Description	Expected Range / Target Value
DIP_01.1	Electrical supply	12 V – 24 V
DIP_01.2	Electrical supply	12 A maximum
DIP_01.3	Electrical supply	Direct current
DIP_03.1	Environment	Laboratory ambient temperature: 0 °C – 35 °C
DIP_03.2	Environment	Laboratory ambient humidity: 20 % – 80 %
DIP_03.3	Project budget	R 6000

3.3 Functional Requirements

In this section, the primary and top-level derived functional requirements (FRs) for the joint design are defined. The only FRs that are considered are associated with joint operation. FRs are categorised into primary and derived FRs, in which:

- Stakeholders are directly concerned with primary FRs.
- Derived FRs are required so that the system can perform primary FRs.

As seen in Table 3.3 below, the FRs for the joint design are listed.

Table 3.3: Joint Functional Requirements

FR ID	Description	Primary / Derived
FR_01	Articulate output link	Primary
FR_02	Hold position	Primary
FR_03	Measure angular displacement	Primary
FR_04	Store angular displacement data	Derived
FR_05	Release position	Primary
FR_06	Cost effective	Derived
FR_07	Readily available components	Derived

In Figure 3.1, the functional requirements of the joint are depicted below.

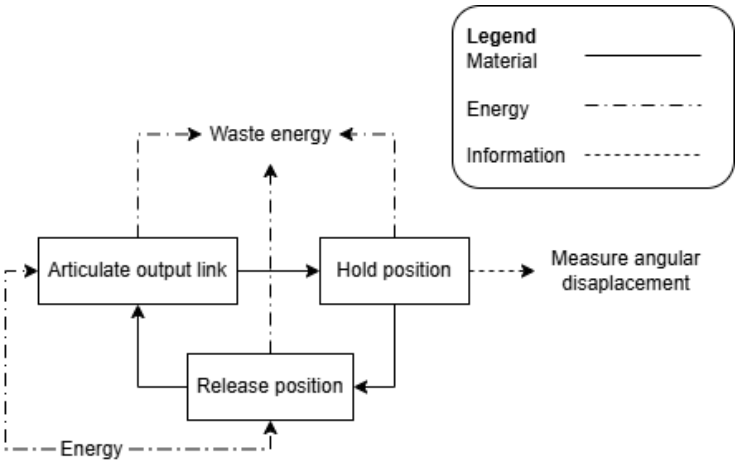


Figure 3.1: Top Level Functional Decomposition

3.4 Technical Performance Measures

This section defines the technical performance measures (TPMs) which will be used to determine how well the joint design (NSoI) performs the required functions[1]. The TPMs, listed in Table 3.4, were derived from the results obtained from the generalised error-propagation model.

Table 3.4: Joint Technical Performance Measures

TPM ID	Related FR	Description	Target Value
TPM_01	FR_01, FR_05, FR_06, FR_07	Operational articulation angle	90°
TPM_02	FR_01, FR_05, FR_06, FR_07	Run-out tolerance	1×10^{-3} rad
TPM_03	FR_02, FR_06, FR_07	Holding torque	7.725 Nm
TPM_04	FR_03, FR_04, FR_06, FR_07	Encoder resolution	12 bits
TPM_05	FR_01, FR_02, FR_05, FR_06, FR_07	Joint life cycle	1×10^{6} cycles

The following subsections, Section 3.4.1 to 3.4.5, provide detailed calculations and mechanical justifications for TPM_01 to TPM_05, respectively.

[1]TPMs are measurable and unambiguous parameters that can be influenced by designing the joint.

3.4.1 TPM_01: Operational Articulation Angle

The 90° articulation TPM is derived from the D-H kinematic model to ensure that the joint mechanism provides the minimum range of motion needed to cover all measurement coordinates within the PCMA working volume. This target value is a baseline of the D-H transformation chain, which guarantees that the complete kinematic path is fulfilled without restricting the arm's reach.

3.4.2 TPM_02: Run-Out Tolerance

Minute bearing play and shaft imperfections can cause the joint to wobble during rotation. If this wobble is not bounded, it can introduce unintentional spatial drift at the probe tip. TPM_02 ensures that the axis maintains alignment and measurement stability.

The TPM is reliant on two parameters: the nominal joint diameter, $d_{\text{joint}} = 20\,\text{mm}$, selected to match commercially available bearings, and the allowable radial bearing deflection defined under ISO 492:2014 as $\delta r_{\text{bearing}} = 0.01\,\text{mm}$ (12). This deflection represents the standard manufacturer precision tolerance for radial internal clearance and run-out. These parameters are used to calculate the run-out tolerance as shown in Equation 3.1.

$$\begin{aligned} \Theta_{\text{runout}} &= \frac{\delta r_{\text{bearing}}}{\left(\frac{d_{\text{joint}}}{2}\right)} \\ &= \frac{0.010}{\left(\frac{20}{2}\right)} \\ &= 1\times10^{-3}\,\text{rad} \end{aligned} \tag{3.1}$$

By limiting the run-out tolerance, joint rotation only tracks single-axis motion without off-axis tilt, preserving D-H kinematic model integrity and minimising spatial errors at the probe tip.

3.4.3 TPM_03: Holding Torque

If the position-holding mechanism is unable to produce sufficient static holding torque, the weight of the distal links can cause the arm to slip. By establishing TPM_03, the joint is guaranteed to secure and maintain its angular position under a maximum gravity-induced cantilever moment without unexpected drift.

Figure 3.2 depicts the free-body diagram used to calculate the holding torque.

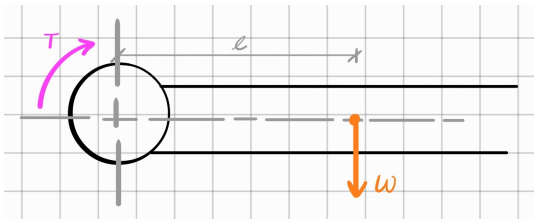


Figure 3.2: FBD of one of the Faro QuantumX links

From the user manual (10), the mass of the link is 1.5 kg, and the distance from the center of the link to the center of the joint is 0.35 m. The holding torque was determined using Equation 3.2, with a safety factor of 1.5 applied:

$$\begin{aligned} T &= SF \cdot m \cdot g \cdot l \\ &= (1.5)(1.5)(9.81)(0.35) \\ &= 7.725 \text{ Nm} \end{aligned} \tag{3.2}$$

By achieving this holding torque, the position-holding mechanism will successfully support the distal links when the mechanism is actuated.

3.4.4 TPM_04: Encoder Resolution

Accurate spatial measurement in a PCMA depends on the ability of each joint's rotary encoder to digitise small angular displacements. If the optical encoder resolution is too coarse, quantisation error is emphasised, preventing the system from meeting uncertainty limits. Establishing TPM_04 ensures that the chosen encoder resolution provides sufficient quantisation to keep measurement error minimised.

The required encoder resolution is calculated using the worst-case uncertainty established under Section 2.2, specifically $\sigma_{\theta,\text{required}} = 1.957\times10^{-5}$ rad. This value comes from the D-H kinematic error propagation derivations, applied across a full rotational range of 360°.

The minimum number of discrete pulse counts, N_{counts}, across a full revolution is calculated using Equation 3.3 below.

$$N_{\text{counts}} = \frac{2\pi}{\sigma_{\theta,\text{required}}} = \frac{2\pi}{1.957 \times 10^{-5}} \approx 320\,983 \text{ CPR} \tag{3.3}$$

This pulse count represents the required encoder resolution to satisfy the worst-case uncertainty. This can be translated into a bit resolution as shown in Equation 3.4.

$$\text{Encoder bit resolution} = \log_2(\text{CPR}) = \log_2(320\,983) \approx 19 \text{ bits} \tag{3.4}$$

A search was conducted to obtain an encoder with the required bit resolution; however, these options were too costly considering the project budget. The highest encoder resolution obtainable within the budget was 1000 PPR, which translates to approximately 12 bits via quadrature decoding.

3.4.5 TPM_05: Joint Life Cycle

By establishing TPM_05, a baseline wear and fatigue threshold is defined to preserve long-term mechanical reliability. This will prevent joint play and measurement repeatability losses over time. Because PCMAs are passive, manually-operated systems rather than motorised, joint lifespans are not constrained by motorised duty cycles. Instead, the baseline requirement, $L_{\text{cycles}} \geq 10^6$ cycles, is established based on industry standards. This ensures that mechanical wear on critical rotating interfaces remains negligible, preserving the validity of the D-H model to ensure accuracy throughout the product's lifecycle.

Chapter 4

Concept Development

4.1 Concept Generation

The TPMs derived during the requirement analysis guided the research into various mechanical solutions, and different mechanisms were evaluated for implementation to ensure the final design could achieve the required TPMs. Three concepts were generated and evaluated against the FRs and the TPMs, as presented in this section.

4.1.1 Concept One: PCMA Joint with a Locking Dial

Figure 4.1 below shows a PCMA joint with the following internal modifications to implement position-holding capabilities. In the center of the shaft is a spacer between two angular contact bearings, which will allow the shaft to rotate with minimal run-out. Bellville washers and preload nuts are used to secure the bearings in place. On one end, the rotary encoder's shaft is connected to the joint's shaft via a rigid coupling and the encoder's body is held secure by the joint housing. This is done to ensure that the relative angular displacement of the shaft with respect to the joint housing is measured. The joint housing is held stationary by the base link and the joint shaft is rotated by the output link. The knob on the other end of the shaft is meant to loosen and/or tighten the preload nuts against the bearings to lock or release the shaft for the output link's rotation.

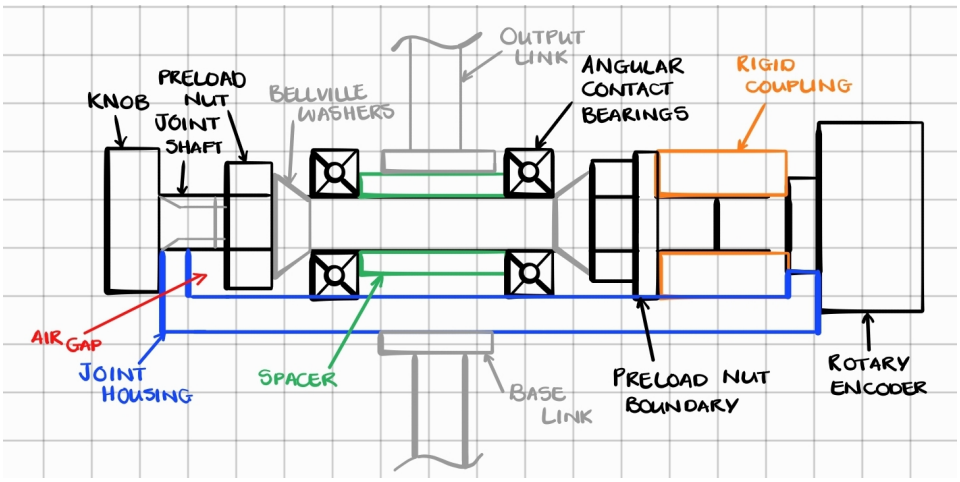


Figure 4.1: PCMA Joint with Locking Dial

4.1.2 Concept Two: PCMA Joint with a Counterbalance Spring

Figure 4.2 illustrates a PCMA joint with external modifications to implement position-holding. The counterbalance spring produces a lifting force to the upper link via the upper and lower rods. Each rod is pin-connected to a cuff that can be fixed in position along the length of either link using a set screw, thereby holding the position of the joint. The spring between the upper and lower rods is used to increase the range of rotation for the joint.

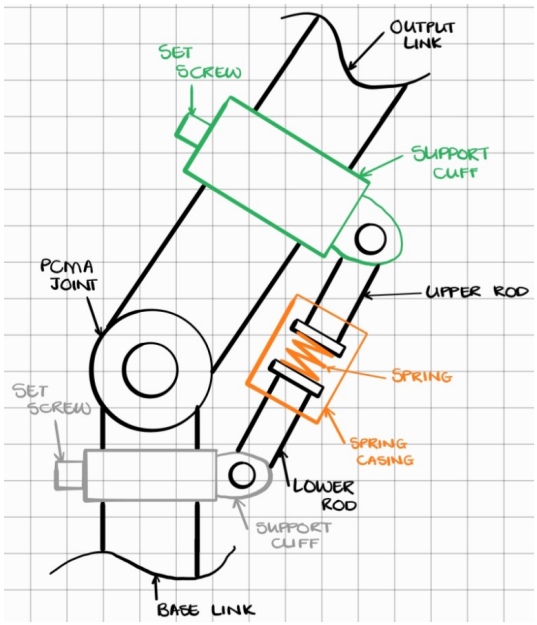


Figure 4.2: PCMA Joint with Couunterbalance Spring

4.1.3 Concept Three: PCMA Joint with an Electromagnetic Brake

Figure 4.3 depicts a PCMA with internal modifications to implement an electromagnetic brake. In the center of the shaft, two angular contact bearings are fitted in place with the bearing housing and pressed against a shoulder of the shaft by preload nuts and washers. On one side of the shaft, the rotary encoder's shaft is connected to the joint's shaft via a rigid coupling and the encoder's body is held stationary by the joint housing. The base link is supported by the joint housing. The electromagnetic brake sits on the other side of the shaft and is also held stationary by the joint housing, however, the brake disc is free to rotate on the joint shaft. When the brake is energised, the air gap between the brake disc and friction plate opens up and the shaft is free to be rotated by the output link. When the brake's power supply is cut off, the air gap closes and the joint is held still.

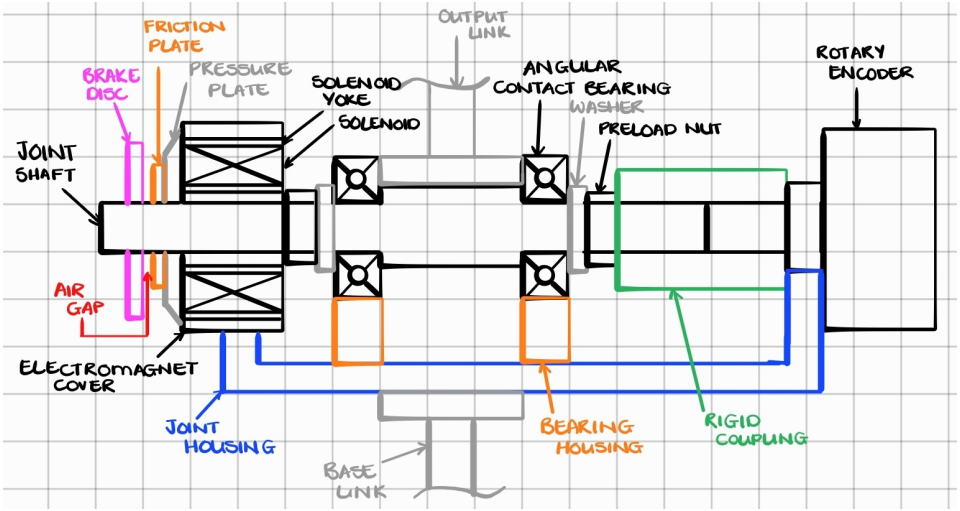


Figure 4.3: PCMA Joint with Electromagnetic Brake

4.2 Concept Evaluation

The concepts presented in the previous section were determined to be feasible, therefore, a qualitative evaluation of the concepts is to be conducted for a full quantitative comparison. This section compares the concepts in terms of the following evaluation criteria (EC):

EC1 Load capacity: The ability of the joint design to carry the required holding torque (TPM_03).

EC2 Operational articulation angle: The ability of the joint design to achieve the full required sweep without interference (TPM_01).

EC3 Positional stability: The ability of the joint design to maintain position when the position-holding technology is in use (TPM_02).

EC4 Component availability: The degree to which all required components are available in the mechanical workshop (TPM_04, TPM_05).

EC5 Construction simplicity: The number of distinct components and the ease of assembly and alignment during fabrication (TPM_04, TPM_05).

EC1 was derived from TPM_03 and determined to be the most important EC as the joint will be ineffective if it is unable to achieve the holding torque dictated by TPM_03. EC2 was derived from TPM_01 as the chosen joint design must not inhibit the PCMA's existing range of motion once implemented. EC3 was derived from TPM_02 as any internal sources of error must be minimised to maximise the joint design's ability to maintain and/or improve the PCMA's accuracy. EC4 and EC5 was included as a practical design consideration given the manufacturing capabilities of the mechanical workshop and the project's timeline and budget in consideration of TPM_04 and TPM_05 (FR_06 and FR_07).

Table 4.1 below, presents a comparison of the generated concepts. To rate the concepts, one of them is marked as the DATUM and the other two are awarded a $+1$ if the other concept is stronger or -1 if it is weaker. Each rating is motivated in the table as well.

Table 4.1: Concept Evaluation

Criterion			**Concept 1: PCMA Joint with a Locking Dial**		**Concept 2: PCMA Joint with a Counterbalance Spring (DATUM)**		**Concept 3: PCMA Joint with an Electromagnetic Brake**	
#	**Description**	**Wt.**	**Rtg**	**Motivation**	**Dat**	**Motivation**	**Rtg**	**Motivation**
EC1	Load capacity	5	-1	The capability of the design achieving the exact holding torque required is dependent on the operator's strength and consistency, otherwise the joint design is susceptible to slipping under load.			+1	An apprporiately sized electromagnetic brake can deliver a consistent, static holding torque that meets the FR_02. This removes human reliability (in comparison with concept 1) and ensures reliable load capability when the joint is powered.
EC2	Operational articulation angle	5	+1	The locking mechanism is entirely internal. Because there are no external linkages, the joint can achieve its full required 90° sweep without any physical interference.			+1	Similarly to concept 1, the electromagnetic brake is neatly integrated with the joint shaft, which allows for uninhibited articulation

Table 4.1 – *Continued from previous page*

Criterion			Concept 1: PCMA Joint with a Locking Dial		Concept 2: PCMA Joint with a Counterbalance Spring (DATUM)		Concept 3: PCMA Joint with an Electromagnetic Brake	
#	**Description**	**Wt.**	**Rtg**	**Motivation**	**Dat**	**Motivation**	**Rtg**	**Motivation**
EC3	Positional stability	5	-1	Manually twisting the dial introduces asymmetric axial and radial loads onto the joint shaft. This can cause deflections that violate the FR_01 and FR_05 and degrade the PCMA's spatial accuracy.			+1	The electromagnetic brake applies a uniform, symmetric clamping force between the brake disc and friction plate. This eliminates manual influence during the locking process and minimises parasitic deflection and protects the system's positional accuracy.
EC4	Component availabilty	5	+1	The required components (threaded shafts, preload nuts, bellville washers) are items that either readily available or easily machined in the university workshop.			0	While electromagnetic brakes are commercially available, specific ordering is required, lead times, and budget allocation compared to the raw materials used in the datum design.
EC5	Construction simplicity	5	0	The mechanical assembly is relatively straightforward and on the level of simplicity as the datum design. However, achieving accurate bearing alignment under varying manual requires extra care.			-1	This is the most complex design to implement. It requires precise mechanical calibration of the encoder, as well as integration of electrical wiring and control circuitry for the solenoid.
Number of + ratings			2				3	
Number of - ratings			2				1	
Overall total			0				2	
Weighted total			0		0		+10	

4.3 Concept Selection

Based on the qualitative evaluation presented under Section 4.2., Concept 3 is selected as the final design, as shown in Figure 4.1, under Section 4.1.3.

The motivation behind selecting this concept is its superior performance across the most critical functional criteria, specifically load capacity (EC1) and positional stability (EC3).

The electromagnetic brake provides a consistent and reliable static hold, in contrast to the manual locking dial which relies heavily on human reliability and strength. When de-energised, the electromagnetic brake is able to successfully meet the strict holding force requirement by applying a uniform clamping

force between the brake disc and friction plate. The uniform clamping eliminates off-center loads and minimises parasitic deflections, thereby preserving tight run-out tolerances required for the PCMA's spatial accuracy. In addition to this, the electromagnet is intergrated internally on the joint shaft, ensuring that the joint design does not impede the joint's physical movement. This allows the arm to achieve the required 90° operational articulation sweep without interference (EC2).

To confirm that the chosen selected concept is well suited to perform the functional requirements in Table 4.1, Table 4.2 below, identifies the components to fulfilling the respective functions.

Table 4.2: Concept Functional Mapping

Function Number	**Function Description**	**Concept**
FR_01	Articulate output link	Energising the electromagnetic brake opens the air gap, allowing the joint shaft to be freely rotated by the output link.
FR_02	Hold position	When the electromagnet is de-energised, the air gap closes and the mechanism applies a static holding force to keep the joint stationary.
FR_03	Measure angular displacement	A rotary encoder is connected to the joint shaft via a rigid coupling, measuring relative rotation while the encoder body is held stationary by the housing.
FR_04	Store angular displacement data	The angular displacement data captured by the rotary encoder is transmitted to the microcontroller, which is part of the system's switching circuitry.
FR_05	Release position	Energising the brake's solenoid retracts the pressure plate from brake disk, releasing the joint for movement.

4.4 Design Refinement

The conceptual design of the electromagnetic joint was refined into a 3D CAD assembly to verify spatial feasibility and integration into the wider system of

interest, as seen in Figure 4.4.

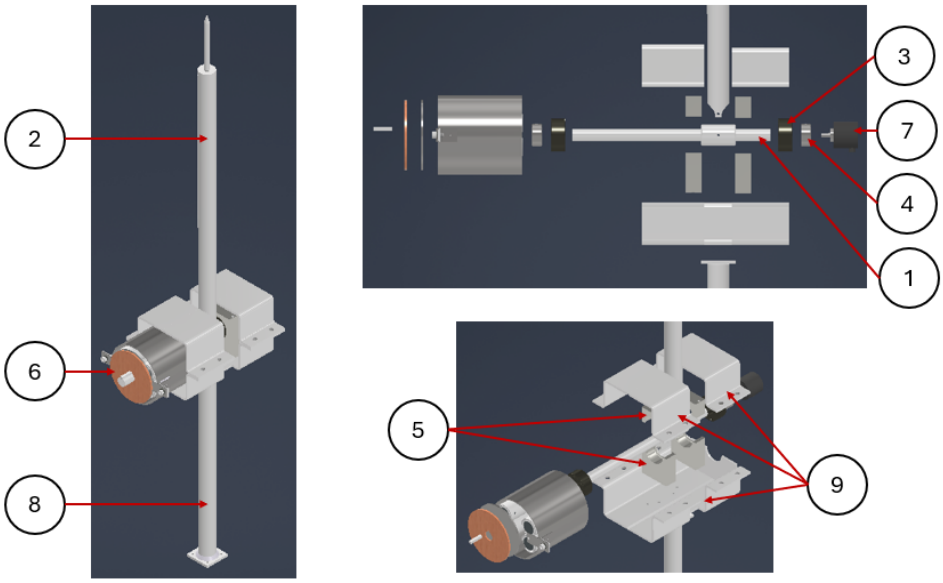


Figure 4.4: Selected Concept Design

The design was modelled to ensure that the main components are correctly sized and positioned to integrate seamlessly within the PCMA's physical constraints. The model also ensures that the means of operation and load paths are clearly defined. This ensures the design does not impede the required 90° operational articulation sweep.The primary mechanical requirement for the refined mechanism is to deliver a relaible static holding torque of 7.725 Nm.

To achieve this, the electromagnetic brake is held stationary by the joint housing, with a loose fit on the joint shaft. The brake disc is fixed to the rotating shaft via a key and the friction plate is connected to the pressure tabs on the elctromagnetic brake and is held stationary as a result. Under normal operation, the electromagnet exerts a force on the friction, which retracts friction plate via the pressure tabs in the electromagnetic brake assembly. This creates an air gap between the brake disc and friction plate to allow the shaft to rotate freely about its bearings. The shaft is rotated by a lightweight output link[1], which is connected to the shaft through a dowel pin.

The position-holding capabilites of the brake are activated by shutting off power to the electromagnet. The pressure tabs return to their resting position and force the friction plate against the brake disc. This is how the umiform clamping force is physically implemented to stop the shaft from rotating, thereby holding the joint's position. This configuration also ensures uniform application of the holding torque to minimise parasitic radial deflections.

The design also accomodates the necessary sensing and structural support subsystems. A high-resolution rotary encoder is connected to the opposite end

[1]The output link represents the distal link of the PCMA arm.

of the joint shaft via a rigid coupling. The encoder's body is supported by the joint housing, which encases the joint assembly and held stationary by the base link[2]. This ensures that any angular displacement of the shaft, relative to the stationary joint housing, is accurately captured and transmitted to the control circuitry. The shaft is supported by two roller bearings, which were chosen over angular contact bearings due to their cost-effectiveness and market availability in comparison. These bearings are mounted onto the shaft with an interference fit and shaft collars to prevent axial slipping. In addition to this, the shaft was designed with shoulders to maintain spacing, while the bearing housing was integrated into the joint housing to further eliminate axial play.

Below is a parts list that corresponds to the refined joint subsystem in Figure 4.4, excluding standard/trivial fasteners:

Table 4.3: Refined Design Parts List

Item No.	**Component Description**	**Quantity**
1	Joint shaft	1
2	Output link	1
3	Roller bearings	1
4	Shaft collars	2
5	Bearing housing	2
6	Electromagnetic brake assembly	2
7	Rotary encoder	1
8	Base link	1
9	Joint housing	1

[2]The base link represents the proximal link of the PCMA arm.

Chapter 5

Design Analysis

The design analysis of the electromagnetic brake joint is divided into four hierarchical layers, including the WSoI which encompasses the testing enviornment and the PCMA arm, the NSoI representing the joint, its functional subsystems, and the individual components. The physical development of the joint was iterative, however, the resulting architecture is documented in a logical, top-down fashion instead of a chronological sequence. To describe the system's integration, the analysis is split into three domains which constitute the mechanical design, the electronic design, and the software design. The mechanical design outlines the physical configuration and loading paths. The electronic design describes the peripheral circuitry and power interfacing. The software design defines the control logic and the state machine controlling the electromagnetic brake.

5.1 Mechanical Design

Figure 5.1 shows the 3D CAD assembly of the electromagnetic brake joint produced as part of the detailed design analysis following the integration and refinement of the subsystem concepts.

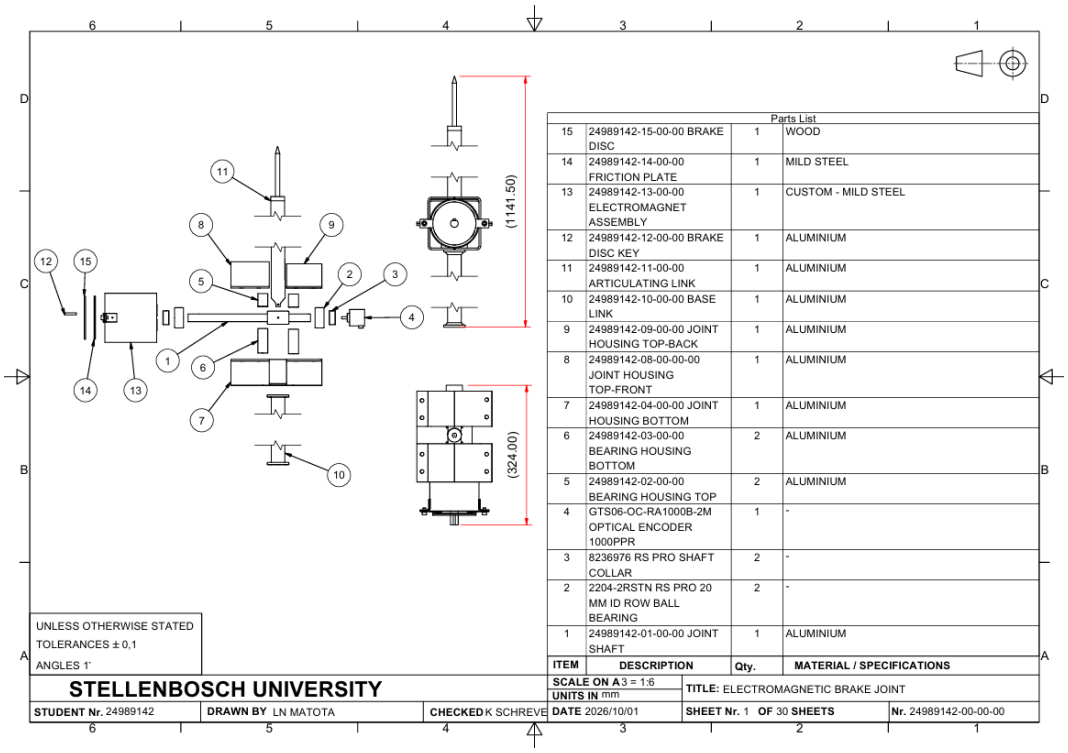


Figure 5.1: Electromagnetic Brake Assembly

The top-level system elements are listed in the drawing's parts list and are marked with balloon item numbers in Figure 5.1.The engineering drawings in Appendix C detail the physical mounting interfaces with external structural components, as well as the spatial clearance limits defined for the joint mechanism within the NSoI.

Material selection for the joint assembly is governed by two competing engineering requirements, which include minimizing moving inertia and maximizing magnetic performance:

- An aluminium alloy, readily available in the workshop, was chosen for the structural housing, joint shaft, bearing blocks, base link, and articulating link. With an assumed density of $\rho_{\text{Al}} = 2700\,\text{kg/m}^3$, aluminium provides a high strength-to-weight ratio that reduces the overall weight of the extended joint parts and minimises the twisting force caused by its own gravity. This enables precise manual arm manipulation without excessive user fatigue.

- A mild steel alloy, also readily availble in the workshop, was chosen specifically for the electromagnet yoke, solenoid cores, pressure tabs, and friction plate. Ferromagnetic mild steel exhibits high magnetic permeability ($\mu_r \approx 2000$) and a high saturation flux density ($B_{\text{sat}} \approx 1.6\,\text{T}$). This allows the magnetic field lines to flow easily and densely through the steel component. It is also important to notes that, its low coercive force minimizes

residual magnetism (B_r) upon power cutoff, which ensures immediate mechanical disengagement when the brake is de-energized.

- A hardwood disc with an assumed $\mu = 0.6$ mating against mild steel, provides a high coefficient of static friction without cold-welding, while a brass key transmits torque to the shaft without inducing magnetic leakage.

The overall configuration uses the designed electromagnetic brake assemblymounted coaxially on the joint shaft with the rest of the joint components. The joint shaft is supported by two RS PRO 20 mm ID double-row ball bearings mounted with an interference fit and secured with RS PRO 32 mm OD one-piece steel shaft collars to prevent axial slipping. The bearing housing are integrated directly into the stationary joint housing. The joint housing clamps down onto the electromagnetic cover and holds the rotary encoder body stationary. The rotary encoder, a GTS06-OC-RA1000B-2M optical encoder (1000 PPR) is fitted into a centered axial borehole and clamped via a grub screw onto the encoder shaft flat. This allows the encoder shaft to rotate with the joint shaft, to measure relative angular displacement (refer to Drawing 24989142-01-00-00 in Appendix C). On the opposite end, the electromagnetic brake assembly directly interfaces with the shaft. The assembly consists of the four solenoid yokes, four solenoid coils, a welded, mild-steel cover, laser-cut mild steel friction plate (laser-cut by Fabrinox), and wooden brake disc.

The mechanical load path originates at the articulating link assembly ($m_{\text{link}} = 0.4659$ kg, including the link pipe, probe hub, and measuring probe). Gravity acting on this mass at an effective moment arm of $l_{\text{link}} = 0.325$ m generates a static gravitational torque. Accounting for a safety factor of 1.5, the actual verified link holding torque is calculated as $T_{\text{holding, verified}} = 2.228$ N·m (Equation B.1 in Appendix B.1). Because this holding torque is within bounds of the TPM_03, calculations following the determination of T_{holding} will use the initially set 7.725 Nm. The torque is countered across the friction interface between the rotating wooden disc and the non-rotating mild steel friction plate. Operating under assumed uniform pressure with a friction coefficient of $\mu = 0.35$, the required axial clamping force generated across this interface is $F_{\text{clamping}} = 636.88$ N (Equation B.2 in Appendix B.2). When power is cut off, the mild steel pressure tabs deliver a combined electromagnetic clamping force of 3550 N (well exceeding the 383.18 N threshold), pushing the friction plate flush against the brake disc.

This coaxial configuration helps maintain a uniform clamping force distribution that minimizes parasitic shaft deflections. Additionally, housing the brake internally allows the joint to achieve its required 90° operational articulation sweep.

Detailed Free Body Diagrams (FBDs), stress analysis derivations, and full numerical holding force calculations supporting these operational conditions are provided in Appendix B.1.

5.2 Electrical Design

The purpose of the switching circuit is to safely bridge the power gap between the low-power control and data management logic and the high-power electromagnetic brake. The STM32 Nucleo microcontroller operates on a 3.3 V logic level with general purpose input/output (GPIO) pins that supply a maximum current of 25 mA. In contrast, the full-scale electromagnet, from the assembly, requires an operating current of $I_{\text{total}} = 8$ A at $V_{\text{DC}} = 24$ V. A direct connection between these two components would cause catastrophic failure of the microcontroller. As a result, a dedicated switching circuit is required to isolate the logic hardware, protect against inductive transients caused by the electromagnet's actuation, and efficiently switch the high-current coil load.

Figure 5.2 details the physical architecture of the electrical subsystem.

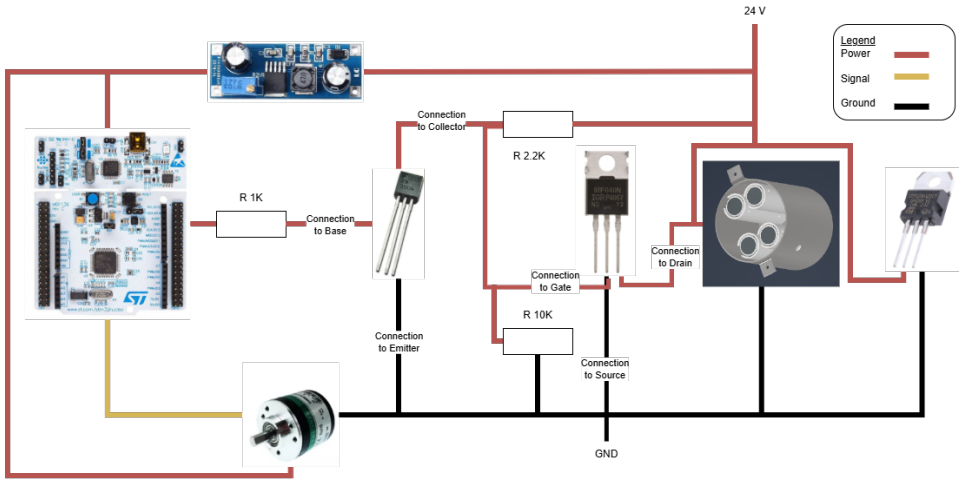


Figure 5.2: Wiring and Interconnect Diagram for the Full-Scale Joint Mechanism

To generate the required total clamping force, four identical solenoids are utilised within the electromagnet. Choosing between a parallel or series electrical interconnection significantly impacts circuit reliability, supply voltage requirements, and coil winding specifications:

- Supply voltage and current distribution: In a series connection, the 24 V supply voltage is divided equally across the four coils (6 V per coil), while the full line current of 8 A flows sequentially through every solenoid. Achieving 8 A at 6 V requires an extremely low coil resistance, demanding thick gauge wire that is difficult to wind densely. As opposed to a parallel configuration, which delivers the full 24 V_{DC} across each solenoid. The 8 A total supply current divides equally among the four parallel branches ($I_{\text{sol}} = 2$ A per coil), resulting in a manageable coil resistance requirement of $R_{\text{coil}} = 12\,\Omega$.

- Redundancy and fault-finding: In a series configuration, an open-circuit

failure in a single solenoid winding disrupts the current path for the entire brake system, leading to complete joint failure. In a parallel configuration, each solenoid operates independently. If one coil open-circuits, the remaining three solenoids can continue to generate holding force.

- Inductive time constant ($\tau = L/R$): A parallel connection maintains high single-coil resistance relative to inductance, which minimises current rise delays during actuation and ensures rapid flux buildup.

Therefore, the parallel configuration was selected for the full-scale joint assembly.

To supply the low-voltage control and feedback hardware from the primary 24 V_{DC} DC supply, an XL7015 DC-DC buck converter is integrated into the power rail. The XL7015 accepts an input voltage range of 0–80 V_{DC} and steps it down to a regulated 5 V output voltage that is suitable for powering the STM32 microcontroller board and the GTS06-OC-RA1000B optical rotary encoder. A high-efficiency DC-DC buck converter ($\eta \approx 85$–90%) ensures stable voltage delivery during high-current load switching transients.

To control the actuation of the electromagnet, an N-channel Power MOSFET (Infineon IRF640N) is deployed in a high-side of the switching circuit acting as a solid-state digital relay.

Because the IRF640N requires a gate-to-source voltage (V_{GS}) of at least 10 V to operate its saturation region, driving its gate directly from the STM32's 3.3 V logic signal would operate the MOSFET in its linear region. This would cause excessive thermal runaway and result in catastrophic damage. To bridge this voltage threshold using components available in the laboratories in the Electrical and Electronics Department, a 2N3904 NPN BJT pre-amplifer stage is integrated into the switching circuit.

The STM32 GPIO signal drives the base of the 2N3904 through a current-limiting resistor of 1 kΩ. When turned off, the 2N3904's collector is pulled up to a 12 V supply rail via a pull-up resistor ($R_{PU} = 2.2$ kΩ), delivering $V_{GS} \approx 9.84$ V directly to the IRF640N gate. Driven to saturation, the IRF640N operates at its rated static drain-to-source resistance $R_{DS(\text{on})} = 0.15\,\Omega$. At the full operating load of $I_{\text{total}} = 8$ A, power dissipation P_{cond} across the MOSFET is determined by Equation 5.1:

$$P_{\text{cond}} = I_{\text{total}}^2 \cdot R_{DS(\text{on})} = (8\text{ A})^2 \cdot 0.15\,\Omega = 9.6\text{ W} \tag{5.1}$$

To dissipate this heat continuously and maintain the junction temperature (T_J) safely below 175°C (as stated in its datasheet), the TO-220 packaged MOSFET is fitted with a dedicated aluminum heatsink.

To maintain the circuit's operability, protect the microcontroller, and ensure reliable gate-driving levels, specific passive components are incorporated as shown in Figure 5.2:

- A base-limiting resistor of 1 kΩ is placed between the STM32 GPIO signal pin and the base of the 2N3904 NPN transistor. Sizing this resistor

restricts the current drawn from the microcontroller GPIO pin during active switching, as seen under Equation 5.2:

$$I_B = \frac{V_{\text{GPIO}} - V_{BE}}{R_B} = \frac{3.3\ \text{V} - 0.7\ \text{V}}{1000\ \Omega} = 2.6\ \text{mA} \tag{5.2}$$

which safely complies with the STM32 absolute maximum output pin threshold of 25 mA.

- A collector pull-Up resistor of 2.2 kΩ is connected between the 24 V DC power supply and the 2N3904 collector node (which connects directly to the IRF640N MOSFET Gate). When the 2N3904 is turned off (GPIO LOW), this resistor pulls the MOSFET Gate voltage up toward the rail, which provides sufficient gate-to-source voltage (V_{GS}) to fully enhance the IRF640N power MOSFET into saturation.

- A gate pull-down resistor of 10 kΩ is connected between the IRF640N MOSFET Gate and Ground (GND). In conjunction with collector pull-down resistor, it forms a voltage divider establishing an active gate drive voltage when the 2N3904 is off, as shown under Equation 5.3:

$$V_{GS} = V_{\text{rail}} \cdot \left(\frac{R_{PD}}{R_{PU} + R_{PD}} \right) = 24\ \text{V} \cdot \left(\frac{10\ \text{k}\Omega}{2.2\ \text{k}\Omega + 10\ \text{k}\Omega} \right) \approx 19.67\ \text{V} \tag{5.3}$$

  This maximizes channel conductivity while remaining safely within the IRF640N maximum absolute gate rating ($V_{GS(\text{max})} = \pm 20$ V). Additionally, during microcontroller boot or floating GPIO states, R_{PD} keeps the gate referenced to Ground.

- An flyback diode is placed in parallel across the electromagnet terminals. When the MOSFET switches off to disengage the joint, the collapsing magnetic field in the coil induces a severe negative Back-electromotive force (Back-EMF) spike ($v_L = -L\frac{di}{dt}$). The flyback diode array provides a means of dissipating the inductive current without exceeding the MOSFET breakdown rating.

For real-time spatial positioning and angular displacement tracking, the GTS06-OC-RA1000B optical rotary encoder (1000 PPR) is connected across the 5 V regulated by the DC-DC buck converter. Its A and B quadrature signal lines are routed directly to the STM32 timer input pins configured in encoder interface mode.

Complete thermal dissipation modeling, component voltage/current distributions, and solenoid winding derivations based on the required holding clamping force are presented in Appendix B.3.

5.3 Software Design

In mechatronics systems, such as the electromagnetic brake system, it is necessary to define the interactions between actuators (electrmoangetic brake), sensors (encoder), and the control software for reliable operation. The software architecture for the brake system uses a supervisory control and data acquisition (SCADA) hierachy to manage real-time monitoring and actuation processes. This hierarchy consists of the following levels, as shown in Figure 5.3:

- Level 0 (field devices) consists of the hardware interacting directly with the optical rotary encoder, the IRF640N MOSFET, and the electromagnet brake.
- Level 1 (control logic) is executed by the STM32 Nucleo microcontroller, which acts as input/output processing module.
- Level 2 (supervisory) is handled by the connected computer that recieves angular displacement data from the microcontroller for logging and visualisation.

To fulfill the system's FRs, the microcontroller's pinout is configured to interface with the Level 0 devices. The rotary encoder's Phase A and Phase B signal lines are connected to pins PA0 and PA1. These pins are routed internally to the STM32's hardware timer (TIM2). The encoder outputs a quadrature pulse[1], which allows the system to determine rotational position and direction of the shaft based on which phase leads the other. By utilising X4 encoding, which is when pulses are counting on the rising and falling edge, for both channels, the 1000 pulse per revolution (PPR) encoder yields 4000 distinct counts per 360-degree rotation.

By operating TIM2 in Encoder Mode, the microcontroller is able to use its dedicated hardware circuitry to autonomously track the high-frequency quadrature pulses and update a 32-bit memory register entirely in the background. This type of tracking eliminates the need for CPU-intensive interrupts, which would force the processor to halt execution of the code to count thousands of pulses per second. As a result, the CPU is freed up to properly manage other tasks. Additionally, the MOSFET gate is driven by pin PC2, which is configured as a digital output. Pin PC13 is configured as a hardware interrupt (EXTI), which is tied to the user input button. Lastly, pins PA2 and PA3 are configured for USART2 serial transmission to communicate the level 2 supervisory computer.

The control strategy implemented for the electromagnetic brake is an open-loop, bang-bang controller. Due to the binary mechanical requirement[2], duty

[1] A quadrature pulse is a signal that consists of two identical square waves offset by 90°.

[2] The joint must be locked in placed, or released, free to rotate.

cycle control is unnecessary. As shown in Figure 5.3, the system initialises into a locked state by defaulting PC2 low, thereby creating a fail-safe protocol.

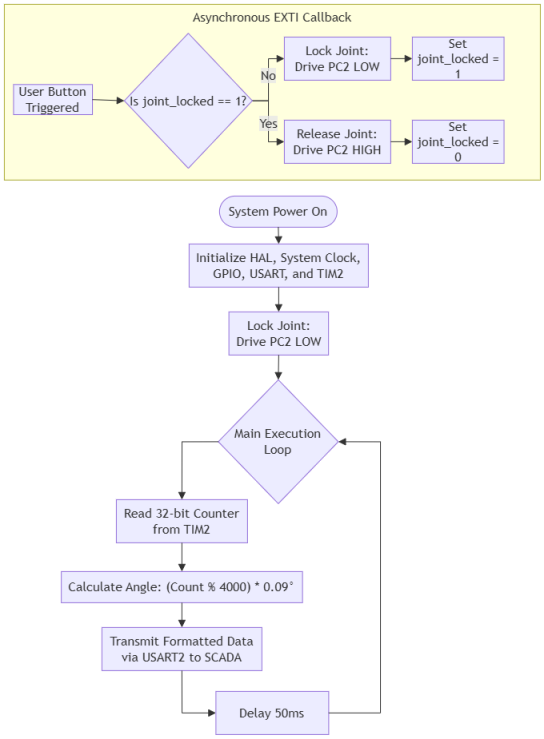


Figure 5.3: Software Design of the PCMA Operation

The main loop reads the 32-bit counter from TIM2, converts the raw pulse count into an angular displacement into degrees, and transmits this data at 20 Hz to the supervisor computer via USART2.

To manage the energisation and de-energisation of the electromagnet, the software uses the external interrupt on PC13. When the user presses the button, the interrupt service routine (ISR) pauses the main looop and toggles the joint's variable (joint_locked) state. If the variable goes high (joint_locked == 1), the controller drives the MOSFET gate high (3.3 V). This energises the solenoid and opens the air gap. Due to the asynchronous execution of the interrupt, the user does not experience lag when manually operating the arm. This is done to ensure a seamless human machine interface (HMI) between the mechanical operation and the software control state.

Chapter 6

Design Evaluation and Testing

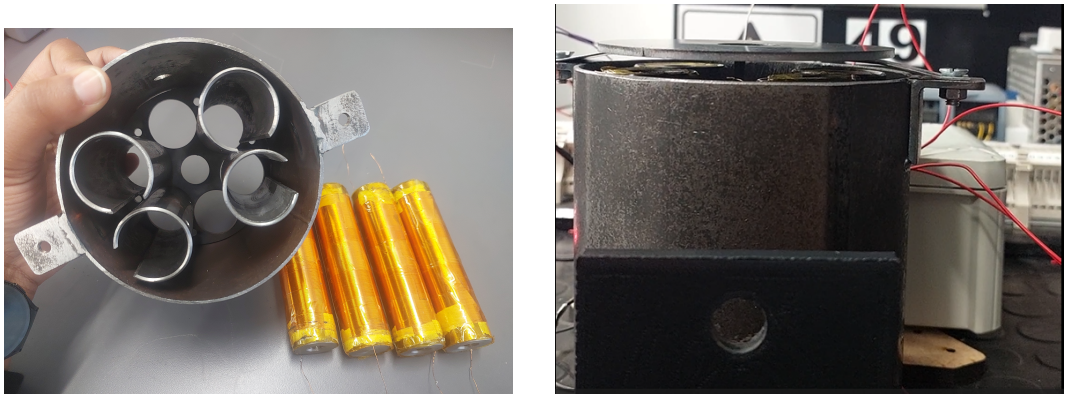
6.1 As-Built Construction

As stated in the project scope, under section 1.3, the construction of the joint design was restricted to critical subsystems required to validate its core operating functions. As a result, instead of building the full-scale joint assembly, only the primary electromagnet with the friction plate was constructed.

Due to project constraints, including an extended theorectical design phase, material availability and manufacturing facilities of the mechanical workshop, the physical build was adapted to the align with these capabilities.

6.1.1 Mechanical Implementation of Electromagnet

The resulting design implementation of the electromagnet resulted in a four-solenoid configuration, wound with enamelled copper wire with a 0.28 mm diameter, as seen in Figure 6.1 below.



(a) Electromagnet Components (b) As-built mechanical housing integration

Figure 6.1: Built Electromagnet Subsystem for Experimental Testing

Due to the project budget and procurement timeline constraints, the appro-

priate copper wire was unattainable and enamelled copper wire with a diameter of 0.28 mm and a maximum operating current of 0.22 A was selected. Each solenoid had two layers of coils spanning a length of at least 10 cm. The winding density was limited to two layers due to the tolerances between the solenoid core and yoke (See Appendix C). This means that the approximate number of coils on each solenoid was determined using Equation 6.1 below:

$$N_{solenoid} = \frac{2L_{solenoid}}{d_{wire}} = \frac{2(100)}{0.28} \approx 715 \text{ coils per solenoid} \tag{6.1}$$

The next step was to evaluate whether the electromagnet could actuate given its reduced capacity. In this case, the force required to deflect the pressure tabs was analysed. Because physical actuation of the electromagnet is dependent directly on the flexure of these tabs, the mechanical assembly was evaluated using the FBD shown in Figure 6.2 below:

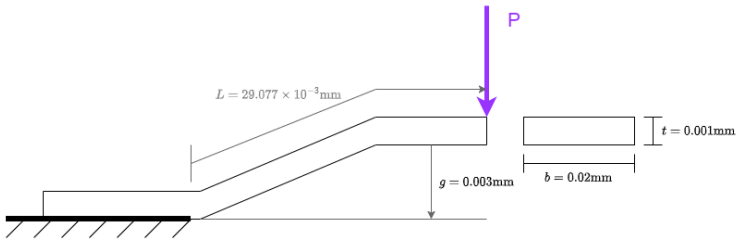


Figure 6.2: FBD to Determine $F_{e(\text{new})}$

Given their structural boundary conditions, the pressure tabs were modelled as cantilever beams. This allows the tip deflection equation $g = \frac{F_{e(\text{new})}L^3}{3EI}$ to be enforced, where g equates to the deflection of the pressure tabs. Assuming a Young's modulus of $E = 210$ GPa for the mild steel tab material, the minimum actuation force $F_{e(\text{new})}$ was calculated using Equation 6.2:

$$F_{e(\text{new})} = \frac{3gEI}{L^3} = \frac{3(0.003)(210 \times 10^9)\left(\frac{1}{12}(0.02)(0.001)^3\right)}{(0.029077)^3} = 128.1 \text{ N} \tag{6.2}$$

This calculated value of $F_{e(\text{new})} = 128.1$ N establishes the revised minimum electromagnetic attraction threshold required to achieve physical brake actuation.

Determining the actual actuation threshold based on mechanical tab deflection rather than the calculated solenoid coil count was necessary due to power uncertainty in the coil windings. While the manufacturer datasheet specifies an absolute continuous operating current limit of 0.22 A, it provides no maximum power dissipation rating. Therefore, the actual I^2R thermal losses within the electromagnet were uncertain. In practice, magnet wire can momentarily sustain significantly higher current levels than its nominal continuous rating during short-duration pulses before thermal breakdown occurs. As will be

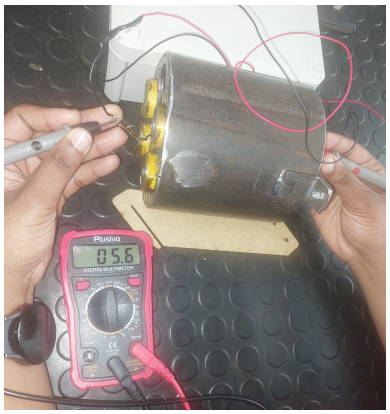
discussed in the electrical implementation section below, this allowable current overrun enabled the electromagnet to be driven well above its nominal datasheet limit to achieve the required actuation force. This shifted the system's primary design constraint from turn count geometry to switching circuit thermal management.

6.1.2 Electrical Implementation of the Electromagnet

The primary objective of the built switching circuit is to safely interface the 3.3 V logic signals from the STM32 Nucleo microcontroller with the high-current electromagnetic brake. The theoretical circuit resistance and impedance were evaluated against physical measurements detailed in Appendix B.4. The measured individual solenoid resistance ($R_{\text{sol}} \approx 17.4\,\Omega$) yields an equivalent parallel array resistance of $R_{\text{eq}} \approx 5.6\,\Omega$, as shown in Figure 6.3 below.



(a) Individual Solenoid Resistance ($R_{\text{sol}} \approx 17.4\,\Omega$)



(b) Total Resistance Across Electromagnet ($R_{\text{eq}} \approx 5.6\,\Omega$)

Figure 6.3: Measured DC Resistance Verification of Solenoid Windings

To provide sufficient current gain ($\beta_{\text{total}} = \beta_1 \cdot \beta_2$) and drive the electromagnetic load from a single microcontroller GPIO pin, an amplifier circuit using a Darlington pair configuration was designed. The stage pairs a small-signal 2N3904 NPN BJT with a high-power TIP41C NPN transistor.

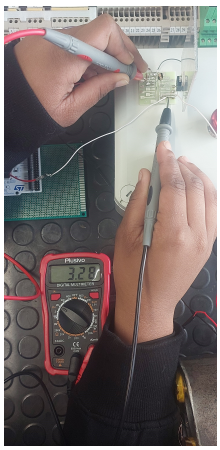
The driver topology was simulated in LTspice prior to physical prototyping. As documented in Appendix B.5, the simulation verified key operational parameters, including:

- Signal Amplification: An input control signal drawn from the STM32 GPIO pin (modelled by the $V1 = 3.3$ voltage supply) effectively controls a load current of $I_C \approx 831.75$ mA through the electromagnet coils (see Figure B.2).
- Collector-Emitter Saturation: The TIP41C power transistor operates deep in saturation, yielding a simulated collector-emitter saturation voltage of

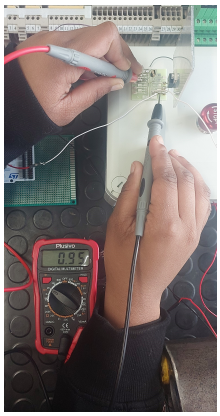
$V_{CE(\text{sat})} \approx 800.72$ mV (0.801 V) at an operational load current of 831.75 mA (see Figure B.4).

- Power Dissipation Sizing: Total power dissipation across the TIP41C transistor during active conduction was simulated at $P_{\text{D(TIP41C)}} \approx 666.82$ mW (0.667 W), confirming that small TO-220 aluminum heatsinks are sufficient to prevent thermal runaway under continuous nominal operation (see Figure B.5).

The measurements seen in Figure 6.4 were recorded prior to connecting the electromagnet to validate the simulation results. The multimeter measurements confirmed an active GPIO logic drive voltage of $V_{\text{GPIO}} = 3.28\,\text{V}$ and a collector-emitter saturation drop of $V_{CE(\text{sat})} = 0.95\,\text{V}$ across the TIP41C. The system was powered via a GW Instek SPD-3606 dual-range DC power supply supplying 0.80 A at 5.80 V during initial stage verification.



(a) GPIO Logic Input Voltage ($V_{\text{GPIO}} = 3.28\,\text{V}$)



(b) TIP41C Saturation Voltage ($V_{CE(\text{sat})} = 0.95\,\text{V}$)



(c) DC Power Supply Output Reading

Figure 6.4: Physical Benchtop Multimeter and Power Supply Verification Readings

6.1.2.1 Heatsinking, Parallel Driver Replication, and Actuation at 4.05 A

During initial testing of the single Darlington switching stage, aluminum heatsinks were mounted to the TO-220 packages of the TIP41C power transistors to assist with thermal dissipation. Although the driver transistors remained cool during low-current energization tests, the electromagnetic field strength generated at the nominal wire limit ($I_{\text{total}} \leq 0.88\,\text{A}$) was insufficient to overcome the bending stiffness of the pressure tabs and achieve physical brake actuation.

To increase the total current delivered to the coil array without exceeding the thermal or collector current ratings ($I_{C(\text{max})} = 6\,\text{A}$) of an individual TIP41C transistor, the switching circuit was replicated and connected in a parallel switching configuration, as seen in Figure 6.5 below.

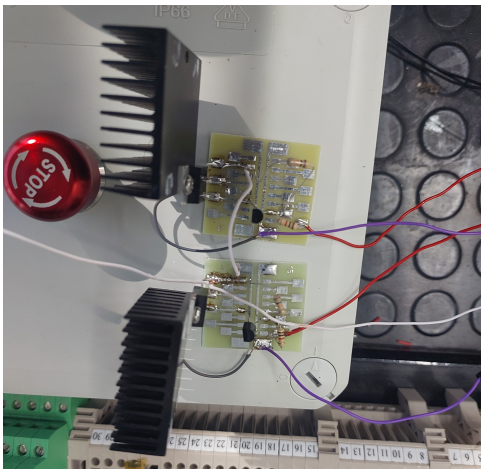


Figure 6.5: Replicated Switching Circuits in Parallel Configuration

This configuration allowed the circuits to share the thermal load and enabled a significantly higher current to be drawn through the electromagnet array.

When the power supply current was progressively increased to $I_{\text{total}} = 4.05\,\text{A}$ (corresponding to a branch current of $I_{\text{sol}} \approx 1.01\,\text{A}$ per solenoid), the generated electromagnetic attraction force initiated visible mechanical clamping of the brake assembly; however, it remained insufficient to achieve full structural deflection of the pressure tabs. A comprehensive evaluation of this mechanical deflection behavior and its implications is presented in Section 6.2.1.

6.1.3 Software Implementation and Duty Cycle Control

The firmware architecture developed for the STM32 Nucleo microcontroller was retained from the main control framework, with data acquisition routines removed due to the absence of the rotary encoder on the electromagnet. To mitigate severe thermal loading and I^2R power dissipation during $I_{\text{total}} = 4.05\,\text{A}$ actuation tests, an intermittent pulse-width control scheme was implemented in software. By configuring extended off-state intervals relative to actuation pulses, the duty-cycled control sequence provided sufficient dwell time for both the solenoid windings and TIP41C power transistors to cool down between test cycles, ensuring safe and repeatable physical operation.

6.2 Safety and Performance Testing

Experimental validation of the electromagnet was conducted in the Mechatronics Laboratory in accordance with Stellenbosch University safety regulations. Prior to testing, a formal risk assessment was executed, covering electrical insulation checks, emergency stop functionality, and high-current driver thermal limit.

6.2.1 Strain Gauge Application and Experimental Setup

To measure structural strain on the pressure tabs and evaluate holding torque, foil strain gauges (350 Ω) were mounted in a full-bridge configuration. The installation procedure followed standard surface preparation guidelines with the use of the strain gauge applicator kit, shown in Figure 6.6 below:

1. A clean glass sheet was utilized as the primary working surface to prevent contamination. The mild steel pressure tabs were degreased with isopropyl alcohol and gause.

2. The mounting zone was conditioned using an acid solution to remove oxide layers, followed by an alkali neutralizer to stabilize the surface pH.

3. Catalyst was applied to the strain gauge backing. The gauge was aligned and bonded using M-Bond 200 cyanoacrylate adhesive, with uniform pressure applied via a Teflon strip for 60 seconds.

4. The celo tape was used as a means of moving the strain gauge from its packaging to the mounting surface.

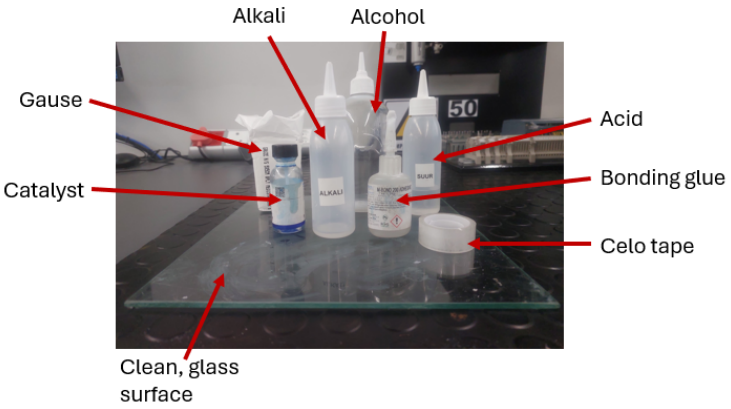


Figure 6.6: Strain Gauge Applicator Kit

The physical test setup connected the electromagnet, the replicated parallel Darlington switching circuit, a GW Instek SPD-3606 DC power supply, an emergency stop button, and a Vishay P3 Strain Indicator and Recorder for signal acquisition, as seen in Figure reffig:screenshot001tsu below.

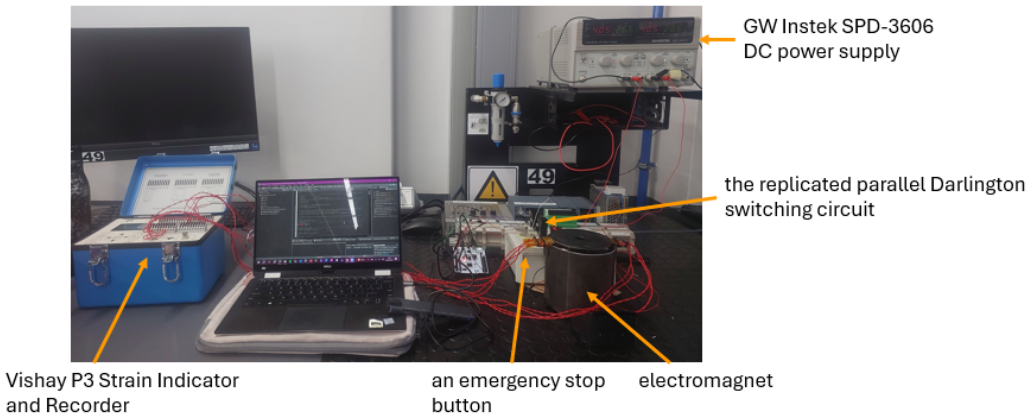


Figure 6.7: Electromagnet Testing Setup

6.2.2 Holding Force Verification

Performance testing was conducted by progressively increasing the current delivered to the electromagnet.

Upon triggering the STM32 GPIO signal, electromagnetic excitation resulted in visible initial movement of the friction plate. However, the generated attraction force was insufficient to achieve the full 3 mm flexure required to completely close the air gap and pull the friction flat firmly against the opening of the electromagnet. Consequently, structural strain could not be transferred to the pressure tabs under electromagnetic actuation alone.

To verify whether the DAQ channel was functioning correctly, manual downward pressure was applied directly onto the pressure tabs. The Vishay P3 display was monitored during manual deflection; however, no change in the initial zero-strain offset (ε_0) was recorded. This confirmed that the lack of strain output during electrical actuation was due to incomplete mechanical deflection across the air gap rather than a signal processing or acquisition fault.

6.2.3 Discussion

The physical testing successfully validated the functional logic of the STM32-driven switching circuit, the signal amplification capability of the 2N3904, and the current-sharing performance of the parallel TIP41C transistors. Under a total current of $I_{\text{total}} = 4.05$ A, the electromagnetic field was strong enough to produce visible mechanical clamping.

However, full physical mechanical actuation (complete closure of the 3 mm air gap) and strain acquisition were limited by a combination of electromechanical and thermal constraints:

- Pressure tab bending stiffness and material elasticity: The required 3 mm deflection distance, combined with the high bending stiffness of the mild

steel pressure tabs, imposed a steep mechanical actuation threshold ($F_{e(\text{new})} = 128.1\,\text{N}$). Utilizing a material with a lower modulus of elasticity ($E < 210\,\text{GPa}$) or a higher elastic yield limit would have significantly reduced the required attraction force, allowing full physical deflection under lower magnetic excitation.

- Solenoid yoke and core clearance: The number of active coil turns was constrained by tolerances between the solenoid yoke and core. The tight inner and outer diameter manufacturing tolerances between the solenoid cores, solenoid yokes, and electromagnet cover restricted winding density to only two layers ($N \approx 715$ turns theoretical maximum). Designing yokes with larger internal diameter tolerances would allow additional coil layers to be added, directly increasing the total magnetomotive force ($\mathcal{F} = N \cdot I$).

- Wire gauge and operating current rating: Budget and timeline constraints necessitated the use of an enamel-coated copper wire with a fine conductor diameter of 0.28 mm and a datasheet continuous rating of only 0.22 A. Using a larger gauge copper wire with a higher continuous current rating would have permitted substantially higher sustained magnetic flux generation without risking thermal degradation.

- Thermal dissipation and circuit component limits: Operating the electromagnet at $I_{\text{total}} = 4.05\,\text{A}$ generated severe I^2R thermal loading ($\approx 305\,\text{W}$ total power dissipation across the coil array). While software duty-cycling and the use of heat-sinks successfully protected the solenoid insulation and TIP41C transistors from thermal breakdown respectively, the short actuation pulses restricted static load testing. Utilizing power transistors with higher thermal and continuous collector current ratings, alongside enhanced active heat sinking, would enable longer continuous duty cycles[cite: 22].

In summary, while full mechanical deflection and static holding torque readings could not be achieved due to the aforementioned constraints, the testing successfully confirmed the core electromechanical operating logic, switching circuit scalability, and safety protocols under laboratory conditions.

Chapter 7

End-of-Life Protocol

To align with modern sustainable engineering principles (ISO 14001) and international directives on Waste Electrical and Electronic Equipment (WEEE Directive 2012/19/EU), a comprehensive End-of-Life (EOL) protocol was established for the electromagnetic brake joint assembly. This protocol outlines design for disassembly, material recovery streams, hazardous substance management, and circular economy strategies.

7.1 Design for Disassembly (DfD) Architecture

The joint assembly incorporates mechanical modularity to ensure rapid non-destructive dismantling at the end of operational life. Threaded fasteners are standardized across the subassemblies to avoid the unnecessary use of permanent adhesive bonds or ultrasonic welds:

1. Removal of external wiring harnesses and detachment of the low-side Darlington switching driver PCB.
2. Unfastening the retaining bolts to release the mild steel yoke, friction plate, and pressure tabs.
3. Sliding the solenoids out of the yokes to disassemble the electromagnet.

7.2 Material Stream Classification and Circular Recycling

Table 7.1 defines the end-of-life material recovery routes for all primary structural and electrical constituents of the joint assembly.

Table 7.1: End-of-Life Material Recycling and Disposal Protocol

Component	Material Group	% Mass	Recommended EOL Recovery Route
Electromagnet Coils	High-Purity Copper (0.28 mm)	32%	Pyrometallurgical / Hydrometallurgical Recycling
Yoke Core & Tabs	1018 Mild Steel Stock	48%	Ferrous Scrap Metal Smelting & Reprocessing
Bobbins & Housing	PETG / Nylon Polymer	12%	Mechanical Polymer Granulating & Regrind
Switching Driver	FR4 Board / Lead-Free Solder	6%	Specialized E-Waste Pyrolysis & Metal Extraction
Insulation / Fasteners	Kapton / Stainless Steel	2%	Material Recovery / Thermal Energy Recovery

7.3 Hazardous Substance Mitigation and Environmental Safety

To comply with Restriction of Hazardous Substances (RoHS Directive 2011/65/EU) guidelines, all electronic components (TIP41C power transistors, 2N3904 BJTs, and 1N4007 diodes) are specified as lead-free and halogen-free.

- The control electronics PCB must be diverted from municipal landfill streams and processed through accredited e-waste facilities to recover trace semiconductor materials.
- Thermal insulation, such as Kapton tape and magnet wire enamel coatings must undergo controlled high-temperature incineration with gas scrubbing to eliminate toxic volatile organic compound (VOC) emissions during metal recovery.
- **Circular Material Efficiency:** The overall recyclable mass fraction of the joint mechanism exceeds 92%, supporting closed-loop manufacturing frameworks for industrial metrology hardware.

Chapter 8

Conclusions

This project successfully developed, modelled, and evaluated a position-holding joint mechanism for a to mitigate operational drift and enhance kinematic stability.

The Denavit-Hartenberg (D-H) kinematic model derived from the FaroArm QuantumX architecture was successfully integrated into a MATLAB-based error-propagation framework. Spatial uncertainty mapping confirmed that position uncertainty is heavily posture-dependent, ranging from a minimum U_{95} spatial uncertainty of 56 μm near the base to a maximum of 74 μm at full extension. This mathematical model proved capable of evaluating single-joint locking scenarios to inform future multi-DOF arm designs.

Based on the error-propagation analysis, TPMs were established, including a static holding torque target of 7.725 N $\cdot$ m (TPM_03) and an operational articulation sweep of 90° (TPM_01). A qualitative evaluation chose an internal electromagnetic brake configuration as the optimal concept, as it provides a uniform clamping force that minimise parasitic deflections while maintaining full articulation range.

The full-scale mechatronic joint design successfully integrated mechanical, electronic, and software domains, specifying an IRF640N MOSFET driven by a 2N3904 NPN pre-amplifier stage to safely switch the 24 V, 8 A parallel coil load from an STM32 Nucleo microcontroller. Real-time angular position tracking would have been achieved via hardware timer encoder interfaces operating at 4000 counts per revolution. For the as-built electromagnet subassembly implementation, a low-side Darlington driver stage (pairing 2N3904 and TIP41C power transistors) was designed, simulated in LTspice, and replicated in parallel to safely validate high-current coil actuation during testing.

Physical testing of the built electromagnet verified the switching logic, current-sharing among parallel Darlington driver stages, and software safety protocols under operational currents up to 4.05 A. Initial mechanical clamping was demonstrated. Although full 3 mm air-gap closure and strain-gauge force readings were constrained by the high bending stiffness of the mild steel pressure tabs, tight solenoid winding tolerances, and thermal limits of the fine wire gauge

(0.28 mm diameter), the experimental apparatus confirmed the overall feasibility of the electromagnetic locking concept.

Overall, the project established a theoretical foundation and partially functional subsystem prototype for position-holding PCMA joints. The generalized D-H error model and standardized design documentation provide a clear framework for future researchers to optimize actuator dimensions and material selection for full-scale arm integration.

Appendix A

Bill of Materials

This appendix provides the itemized Bill of Materials (BOM) for the electromagnetic brake joint assembly. All prices reflect verified vendor quotations, retail receipts, and university workshop billing records inclusive of 15% VAT.

Table A.1: Detailed Bill of Materials (BOM) for Joint Prototype

Category	Part Description	Supplier / Part No.	Qty	Unit Price	Total (ZAR)
Mechanical	Laser-cut Electromagnetic Cover	Fabrinox / 24989142-01-01	5	R 25.81	R 129.05
	Laser-cut Pressure Platform	Fabrinox / 24989142-01-02	5	R 2.38	R 11.90
	Laser-cut Friction Plate	Fabrinox / 24989142-01-03	5	R 9.18	R 45.90
	20mm ID 2-Row Ball Bearings	RS / 2204-2RSTN	2	R 345.29	R 690.58
	Ruland Rigid Shaft Coupling	RS / MCLX-20-20-F	1	R 869.67	R 869.67
	32mm OD Steel Shaft Collars	RS / 8236976	2	R 62.42	R 124.84
	Raw Material Steel / Aluminium	SU Workshop Stock	1	R 3 084.81	R 3 084.81
Electrical	Enamelled Copper Wire (0.28mm)	RS / CUL 100/0,28	4	R 187.66	R 750.64
	Infrared IRF640NPBF MOSFETs	RS / IRF640NPBF	6	R 24.05	R 144.30
	Schottky Diodes STPS20H100CT	RS / STPS20H100CT	6	R 9.12	R 54.72
	Red Hook-Up Wire (4m Reel)	RS / 50747-4	2	R 226.17	R 452.34
	High-Temp Kapton Insulation Tape	Takealot Retail	1	R 299.99	R 299.99
	TIP41C / 2N3904 BJTs & Board	Lab Stock	1 set	R 120.00	R 120.00
Sensors & Power	Optical Encoder (1000 PPR)	Micro Robotics / GTS06	1	R 668.00	R 668.00
	Step-Down Buck Regulator	Micro Robotics / XL7015	1	R 30.00	R 30.00
	Shipping & Delivery Fees	RS / Micro Robotics	1 set	R 567.77	R 567.77
Labor	Workshop Machining (18.5 hrs)	SU Workshop / Paulse	18.5	R 650.00	R 12 025.00
Grand Total					**R 19 977.55**

Appendix B

Design Calculations

This appendix presents the quantitative structural and electromagnetic sizing calculations for the PCMA joint mechanism.

B.1 Articulating Link Mass and Holding Torque Verification

To verify whether TPM_03 remains valid for the finalized joint geometry, the mass of the articulating link assembly, comprising the aluminium link pipe, probe hub, and steel measuring probe, was calculated, under Equation B.1 below. An aluminium density of $\rho_{\text{Al}} = 2700\,\text{kg/m}^3$ and a steel density of $\rho_{\text{steel}} = 7850\,\text{kg/m}^3$ was assumed, using the FBD and safety factor (SF = 1.5) established under Section 3.4.3 (13).

$$\begin{aligned} m_{\text{link}} &= \rho_{\text{Al}} \cdot V_{\text{pipe}} + \rho_{\text{Al}} \cdot V_{\text{hub}} + \rho_{\text{steel}} \cdot V_{\text{probe}} \\ &= \rho_{\text{Al}} \left[\pi(r_o^2 - r_i^2)L_{\text{pipe}} + \pi(r_{\text{top}}^2 + r_{\text{bottom}}^2)L_{\text{hub}}\right] + \rho_{\text{steel}} \left[\pi r_{\text{probe}}^2 L_{\text{probe}}\right] \\ &= 2700 \cdot \left[\pi(0.016^2 - 0.013^2)(0.55) + \pi(0.017^2 + 0.013^2)(0.01)\right] \\ &\quad + 7850 \cdot \left[\pi(0.005^2)(0.1)\right] \\ &= 0.4659\,\text{kg} \\ \therefore T_{\text{holding, verified}} &= \text{SF} \cdot m_{\text{link}} \cdot g \cdot l_{\text{link}} \\ &= (1.5)(0.4659\,\text{kg})(9.81\,\text{m/s}^2)(0.325\,\text{m}) \\ &= 2.228\,\text{N} \cdot \text{m} \end{aligned} \tag{B.1}$$

Although Equation B.1 confirms that the physical link mass results in a required holding torque of 2.228 N·m, the joint brake was conservatively sized against the upper-bound specification requirement of $T_{\text{holding}} = 7.725\,\text{N·m}$.

B.2 Electromagnetic Brake Sizing and Axial Clamping Force

An outer diameter of $d_o = 0.1\,\text{m}$ and inner bore diameter of $d_i = 0.022\,\text{m}$ were chosen for the brake disc and friction plate interface based on the material available in the workshop and mechatronics laboratory, namely the mild steel piping and bars.

Assuming uniform contact pressure across the rigid wood-on-steel friction interface ($\mu = 0.35$), the required axial clamping force F_{clamping} to hold $T_{\text{holding}} = 7.725\,\text{N·m}$ is derived from Shigley's uniform pressure disk brake model (2):

$$\begin{aligned} T_{\text{holding}} &= \frac{2}{3}\mu F_{\text{clamping}} \left(\frac{r_o^3 - r_i^3}{r_o^2 - r_i^2} \right) \\ \Rightarrow F_{\text{clamping}} &= \frac{3 \cdot T_{\text{holding}}}{2 \cdot \mu} \left(\frac{r_o^3 - r_i^3}{r_o^2 - r_i^2} \right) \\ &= \frac{3 \cdot (7.725)}{2 \cdot (0.35)} \left(\frac{0.050^2 - 0.011^2}{0.050^3 - 0.011^3} \right) \\ &= 636.88\,\text{N} \end{aligned} \tag{B.2}$$

B.3 Circuit Sizing and Electromagnet Winding Derivations

B.3.1 Component Voltage, Current, and Power Distribution

Table B.1 details the operating parameters (obtained from the component's respective datasheets) across each node during active steady-state operation at $I_{\text{total}} = 8\,\text{A}$.

Table B.1: Electrical Operating Parameters Across Subsystem Nodes

Subsystem / Component	Node Voltage (V)	Operating Current	Power (W)
STM32 GPIO Signal	3.3 V_{DC}	2.6 mA	8.6 mW
2N3904 Pre-Driver Base	0.7 V_{BE}	2.6 mA	1.8 mW
2N3904 Collector (V_{CE})	0.2 V_{sat}	5.4 mA	1.1 mW
IRF640N Gate (V_{GS})	12.0 V_{DC}	< 1 mA	< 10 mW
Solenoids (4× Parallel)	24.0 V_{DC}	8.0 A (total)	192.0 W
XL7015 DC-DC Converter	24.0 V_{DC}	0.12 A	2.88 W
GTS06 Optical Encoder	5.0 V_{DC}	60 mA	0.30 W
IRF640N MOSFET Drain-Source	1.2 V_{drop}	8.0 A	9.60 W

B.3.2 Electromagnet Winding Design based on Clamping Force

To determine the required turn count (N) per solenoid, the total magnetic force (F_{mag}) generated by the electromagnet is equated directly to the required normal clamping force (F_{clamping}) of the brake.

B.3.2.1 Air Gap Magnetic Flux Density (B)

The magnetic force exerted across the effective cross-sectional area of the electromagnetic brake $A_e = \pi(0.1)^2 = 31.42\times10^{-3}\,\text{m}^2$ with a working air gap $g = 0.3\,\text{mm} = 3\times10^{-4}\,\text{m}$ is expressed by Maxwell's stress tensor from Equation 2.12 under Section 2.4.2.

Solving for the required magnetic flux density B, under Equation B.3:

$$B = \sqrt{\frac{2\mu_0 \cdot \mathcal{F}}{A_e}} = \sqrt{\frac{2 \cdot (4\pi \times 10^{-7}\ \text{H/m}) \cdot 636.88\ \text{N}}{31.42 \times 10^{-3}\ \text{m}^2}} \approx 0.2257\ \text{T} \tag{B.3}$$

B.3.2.2 Turn Sizing (N) per Solenoid

Neglecting iron core reluctance ($\mu_{\text{core}} \gg \mu_0$), the total Magnetomotive Force (MMF) required across the air gap g for a parallel circuit setup is stated under Equation 2.12 under Section 2.4.2, where $\mathcal{F} = F_{\text{clamping}}$.

Therefore, solving for the minimum required turns per solenoid, as seen in Equation B.4:

$$N_{\text{solenoid}} = \sqrt{\frac{\mathcal{F}(2g^2)}{kI_{\text{total}^2}\mu_0 A_g}} = \sqrt{\frac{636.88[2(3\times10^{-3})^2]}{4(8)^2(4\pi\times10^{-7})(\pi(0.015)^2)}} \approx 225\ \text{coils per solenoid} \tag{B.4}$$

Applying a 15% safety factor to compensate for magnetic flux leakage and core reluctance losses:

$$N_{\text{design}} = \lceil 225 \times 1.15 \rceil = 259\ \text{turns per solenoid} \tag{B.5}$$

B.4 Electromagnet Actuation Force and Winding Derivations

B.4.1 Mechanical Actuation Force Threshold ($F_{e(\textbf{new})}$)

Physical actuation of the electromagnetic brake requires deflecting the mild steel pressure tabs by $g = 3\,\text{mm} = 0.003\,\text{m}$. Modeling each pressure tab as a cantilever beam, the required tip deflection force is:

$$F_{e(\text{new})} = \frac{3gEI}{L^3} \tag{B.6}$$

Given Young's modulus $E = 210\,\text{GPa}$, length $L = 0.029\,077\,\text{m}$, width $b = 0.020\,\text{m}$, and thickness $t = 0.001\,\text{m}$, the second moment of area I is calculated as:

$$I = \frac{1}{12}bt^3 = \frac{1}{12}(0.020\,\text{m})(0.001\,\text{m})^3 = 1.667 \times 10^{-12}\,\text{m}^4 \tag{B.7}$$

Substituting these parameters yields the revised minimum actuation force threshold:

$$F_{e(\text{new})} = \frac{3(0.003\,\text{m})(210 \times 10^9\,\text{Pa})(1.667 \times 10^{-12}\,\text{m}^4)}{(0.029077\,\text{m})^3} \approx 128.1\,\text{N} \tag{B.8}$$

Dividing across $N_{\text{sol}} = 4$ solenoids gives the minimum required force per individual coil:

$$F_{\text{sol}} = \frac{F_{e(\text{new})}}{4} = \frac{128.1\,\text{N}}{4} \approx 32.03\,\text{N} \tag{B.9}$$

B.4.2 Air Gap Flux Density (B) and Required Magnetomotive Force ($\mathcal{F}$)

Using Maxwell's stress tensor, the magnetic flux density B required to generate $F_{\text{sol}} = 32.03\,\text{N}$ across an effective pole area $A_p = 1.8 \times 10^{-4}\,\text{m}^2$ is:

$$B = \sqrt{\frac{2\mu_0 F_{\text{sol}}}{A_p}} = \sqrt{\frac{2(4\pi \times 10^{-7}\,\text{H/m})(32.03\,\text{N})}{1.8 \times 10^{-4}\,\text{m}^2}} \approx 0.669\,\text{T} \tag{B.10}$$

Neglecting iron core reluctance ($\mu_{\text{core}} \gg \mu_0$), the Magnetomotive Force ($\mathcal{F}$) required across an air gap $g = 0.5\,\text{mm} = 5 \times 10^{-4}\,\text{m}$ is:

$$\mathcal{F} = N \cdot I_{\text{sol}} = \frac{B \cdot g}{\mu_0} = \frac{(0.669\,\text{T})(5 \times 10^{-4}\,\text{m})}{4\pi \times 10^{-7}\,\text{H/m}} \approx 266.2\,\text{A} \cdot \text{turns} \tag{B.11}$$

B.4.3 Theoretical vs. As-Built Turn Sizing Comparison

The equivalent resistance and inductance parameters of the individual solenoids and the total electromagnet were derived to configure the LTspice simulation. The static DC resistance of a single solenoid is determined by Equation B.12:

$$R_{\text{sol}} = \rho\frac{l_w}{A_w} \tag{B.12}$$

where $\rho = 17.02\,\text{n}\Omega/\text{m}$ is the electrical resistivity of copper at room temperature, l_w is the total length of wire, and A_w is the cross-sectional area of the wire conductor. The self-inductance of a single solenoid is determined by Equation B.13:

$$L_{\text{sol}} = \frac{\mu N_{\text{solenoid}}^2 A_c}{l_c} \tag{B.13}$$

where $\mu = \mu_0\mu_r$ is the magnetic permeability of the mild steel solenoid core (assuming $\mu_r \approx 2000$ for the steel core), A_c is the cross-sectional area of the core, and l_c is the magnetic path length. For calculation purposes, the conductor length l_w and magnetic path length l_c were assumed equal.

The theoretical resistance and inductance values for a single solenoid, as well as the equivalent values for the four-solenoid parallel array, are derived in Equation B.14:

$$R_{\text{sol}} = (17.02 \times 10^{-9})\frac{2\pi(0.015)\left[\frac{2(100)}{0.28}\right]}{\pi(140 \times 10^{-6})^2} \approx 18.61\,\Omega \tag{B.14}$$

$$\mu = \mu_0\mu_r = (4\pi \times 10^{-7})(2000) \approx 2.513{\times}10^{-3}\ \text{H/m} \tag{B.15}$$

$$L_{\text{sol}} = \frac{(2.513 \times 10^{-3})(715)^2\left[\pi(0.015)^2\right]}{2\pi(0.015)\left[\frac{2(100)}{0.28}\right]} \approx 13.49\,\text{mH} \tag{B.16}$$

$$R_{\text{eq,calc}} = \frac{R_{\text{sol}}}{4} = \frac{18.61\,\Omega}{4} = 4.65\,\Omega \tag{B.17}$$

$$L_{\text{eq,calc}} = \frac{L_{\text{sol}}}{4} = \frac{13.49\,\text{mH}}{4} \approx 3.37\,\text{mH} \tag{B.18}$$

B.5 LTspice Circuit Simulations and Experimental Validation

To verify the switching response, component stress limits, and signal amplification of the switching circuit before to hardware implementation, transient analysis simulations were executed in LTspice, as seen by Figure B.1.

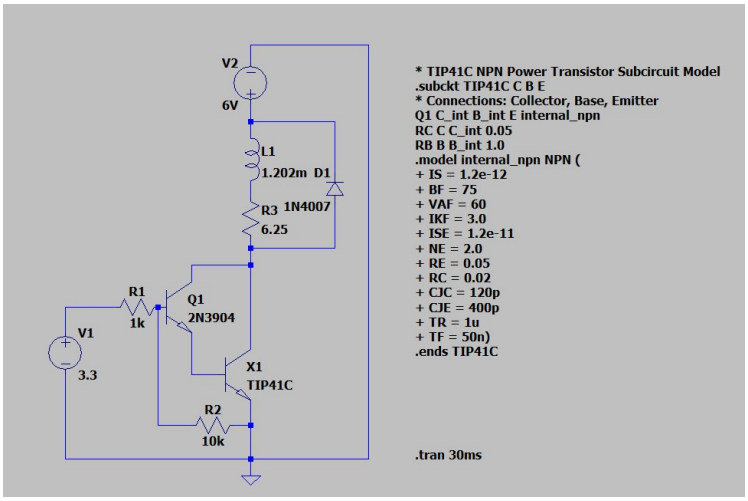


Figure B.1: LTSpice Model

B.5.1 Switching Circuit Component Sizing Derivations

To drive the electromagnetic load from a 3.3 V STM32 GPIO signal without exceeding pin current thresholds, a low-side Darlington driver stage pairing a 2N3904 BJT with a TIP41C power transistor was sized.

B.5.1.1 TIP41C Forced Saturation Sizing

To maintain low collector-emitter saturation voltage ($V_{CE(\text{sat})} \leq 1.0$ V) at a maximum load current of $I_C = 831.75$ mA, a forced saturation gain of $\beta_2 = 15$ was enforced. The required base drive current I_{B2} for the TIP41C is:

$$I_{B2} = \frac{I_C}{\beta_2} = \frac{831.75\,\text{mA}}{15} = 55.45\,\text{mA} \tag{B.19}$$

B.5.1.2 2N3904 and Base Resistor (R_B) Sizing

The 2N3904 amplifies the GPIO input current to deliver $I_{C1} = I_{B2} = 55.45$ mA. Applying a forced gain of $\beta_1 = 20$ for the 2N3904 yields the required GPIO input current:

$$I_{B1} = \frac{I_{C1}}{\beta_1} = \frac{55.45\,\text{mA}}{20} \approx 2.77\,\text{mA} \tag{B.20}$$

The base current-limiting resistor R_B is calculated using Kirchhoff's Voltage Law across the 3.3 V logic signal (V_{GPIO}) and base-emitter drop ($V_{BE1} = 0.70$ V):

$$R_B = \frac{V_{\text{GPIO}} - V_{BE1}}{I_{B1}} = \frac{3.3\,\text{V} - 0.70\,\text{V}}{2.77\,\text{mA}} \approx 938.6\,\Omega \tag{B.21}$$

Selecting a standard nominal resistor of $R_B = 1.0$ kΩ limits the microcontroller pin current to a safe value of $I_{\text{GPIO}} = 2.60$ mA.

B.5.1.3 Transistor Power Dissipation (P_D)

Total continuous power dissipation in the TIP41C during active conduction is calculated as:

$$\begin{aligned} P_{\text{D(TIP41C)}} &= (V_{CE(\text{sat})} \cdot I_C) + (V_{BE2} \cdot I_{B2}) \\ &= (0.801\,\text{V})(0.832\,\text{A}) + (1.10\,\text{V})(0.055\,\text{A}) \approx 0.727\,\text{W} \end{aligned} \tag{B.22}$$

This confirms that continuous thermal dissipation stays well within safe limits when fitted with standard TO-220 aluminum heatsinks.

B.5.2 Signal Amplification and Collector Current Response

Figure B.2 illustrates the current amplification from the microcontroller GPIO signal input (modelled as the V1 voltage supply) to the electromagnet load. The

input signal current $I(V1)$ remains below 1 mA, while the switching transistor delivers a steady load current of $I(R3) = 831.75$ mA through the solenoid array.

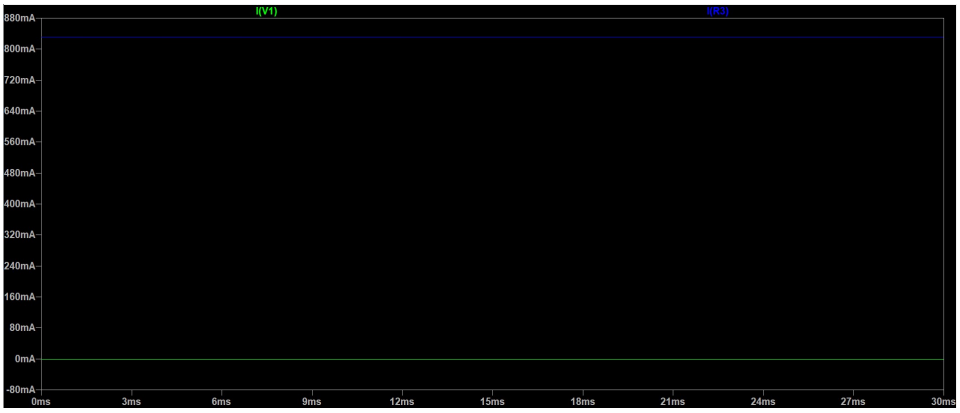


Figure B.2: LTspice Simulation: Input GPIO Signal Current $I(V1)$ vs. Amplified Solenoid Load Current $I(R3) = 831.75$ mA

B.5.3 Inductive Switching Response and Coil Current Decay

Figure B.3 demonstrates the transient current behavior $I(L1)$ across the electromagnet coil inductance. Upon turn-off, the freewheeling 1N4007 flyback diode provides a safe recirculation path for the collapsing magnetic field energy, clamping inductive Back-EMF spikes and protecting the TIP41C collector-emitter junction.



Figure B.3: LTspice Simulation: Inductive Coil Current Transient $I(L1)$ During Switching Operations

B.5.4 Transistor Collector-Emitter Saturation Voltage (V_{CE})

Figure B.4 details the collector-emitter voltage $V(n003)$ and collector current $I(R3)$ across the TIP41C power transistor. At a cursor timestamp of $t = 26.56$ ms, the simulation confirms full saturation with $V_{CE} = 800.72$ mV (0.801 V) at

$I_C = 831.75\,\text{mA}$. This aligns closely with the physical benchtop multimeter measurement of $V_{CE(\text{sat})} = 0.95\,\text{V}$ recorded in Figure 6.4b.



Figure B.4: LTspice Simulation: TIP41C Saturation Voltage $V_{CE} = 800.72\,\text{mV}$ and Collector Current $I_C = 831.75\,\text{mA}$ at $t = 26.56\,\text{ms}$

B.5.5 Power Dissipation in TIP41C Power Transistor

Figure B.5 evaluates total instant power dissipation across the TIP41C transistor, calculated as $P_D = V_{CE} \cdot I_C + V_{BE} \cdot I_B$. The continuous power dissipation settles at $P_{\text{D(TIP41C)}} \approx 666.82\,\text{mW}$ (0.667 W) during steady-state conduction. This verifies that operating the driver continuously at nominal current will not exceed the package thermal threshold when fitted with standard TO-220 aluminum heatsinks.



Figure B.5: LTspice Simulation: Active Power Dissipation Trace across TIP41C Transistor ($P_D \approx 666.82\,\text{mW}$)

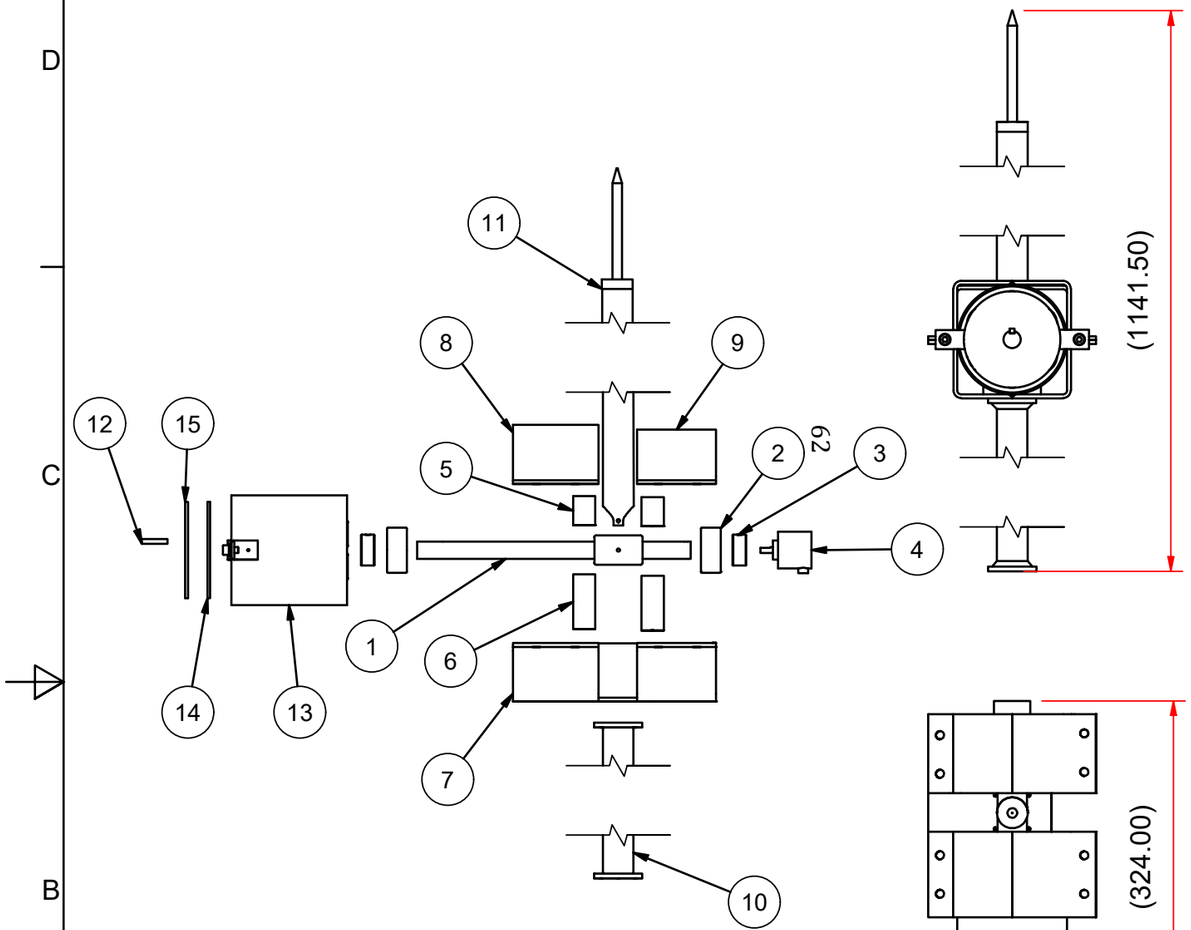
Appendix C

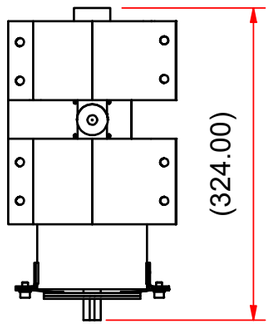
Drawing Pack

Table C.1 presents the hierarchical drawing tree and document breakdown for the Portable Coordinate Measurement Arm (PCMA) electromagnetic joint assembly drawing pack.

Table C.1: Hierarchical Drawing Tree and Document Index for the PCMA Joint Assembly

Drawing Number	Title / Description	Level	Material / Spec.	Qty
24989142-00-00-00	**Electromagnetic Brake Joint Assembly**	**0**	**Assembly**	**1**
24989142-01-00-00	Joint Shaft	1	Aluminium	1
24989142-02-00-00	Bearing Housing Top	1	Aluminium	2
24989142-03-00-00	Bearing Housing Bottom	1	Aluminium	2
24989142-04-00-00	Joint Housing Bottom Assembly	1	Aluminium	1
24989142-05-00-00	Joint Housing Top-Front	1	Aluminium	1
24989142-06-00-00	Joint Housing Top-Back	1	Aluminium	1
24989142-07-00-00	Base Link Assembly	1	Aluminium	1
24989142-08-00-00	Articulating Link Assembly	1	Aluminiumy	1
24989142-09-00-00	Brake Disc Key	1	Brass	1
24989142-10-00-00	Electromagnet Assembly	1	Custom	1
24989142-11-00-00	Pressure Tab Detail	1	Mild Steel	1
24989142-12-00-00	Friction Plate	1	Mild Steel	1
24989142-13-00-00	Brake Disc	1	Wood	1





Parts List			
15	24989142-15-00-00 BRAKE DISC	1	WOOD
14	24989142-14-00-00 FRICTION PLATE	1	MILD STEEL
13	24989142-13-00-00 ELECTROMAGNET ASSEMBLY	1	CUSTOM - MILD STEEL
12	24989142-12-00-00 BRAKE DISC KEY	1	ALUMINIUM
11	24989142-11-00-00 ARTICULATING LINK	1	ALUMINIUM
10	24989142-10-00-00 BASE LINK	1	ALUMINIUM
9	24989142-09-00-00 JOINT HOUSING TOP-BACK	1	ALUMINIUM
8	24989142-08-00-00-00 JOINT HOUSING TOP-FRONT	1	ALUMINIUM
7	24989142-04-00-00 JOINT HOUSING BOTTOM	1	ALUMINIUM
6	24989142-03-00-00 BEARING HOUSING BOTTOM	2	ALUMINIUM
5	24989142-02-00-00 BEARING HOUSING TOP	2	ALUMINIUM
4	GTS06-OC-RA1000B-2M OPTICAL ENCODER 1000PPR	1	-
3	8236976 RS PRO SHAFT COLLAR	2	-
2	2204-2RSTN RS PRO 20 MM ID ROW BALL BEARING	2	-
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ITEM	**DESCRIPTION**	**Qty.**	**MATERIAL / SPECIFICATIONS**

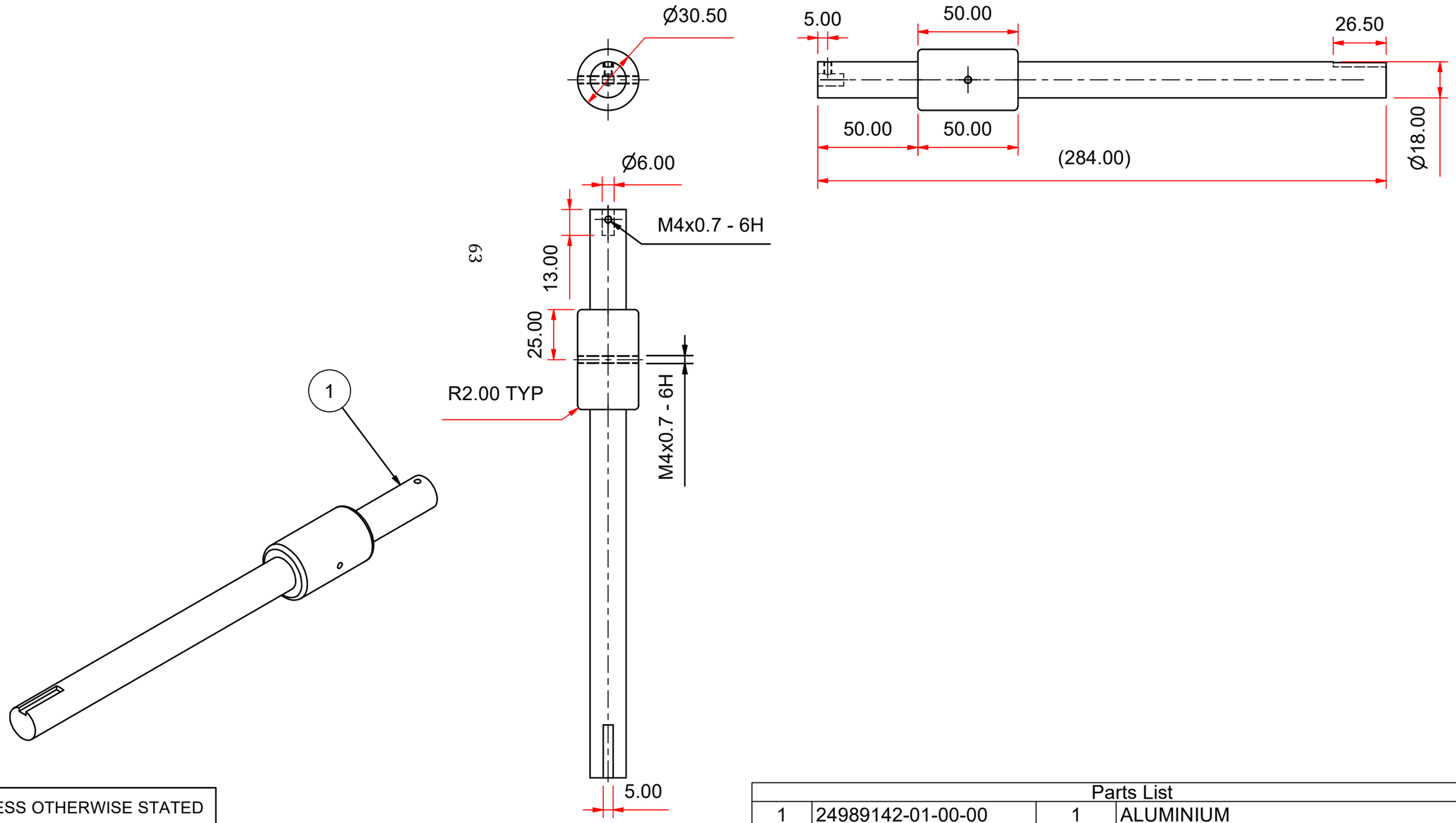
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TOLERANCES ± 0,1
ANGLES 1°

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SCALE ON A3 = 1:6
UNITS IN mm

TITLE: ELECTROMAGNETIC BRAKE JOINT

STUDENT Nr. 24989142 | **DRAWN BY** LN MATOTA | **CHECKED** K SCHREVE | **DATE** 2026/10/01 | **SHEET Nr.** 1 **OF** 30 **SHEETS** | **Nr.** 24989142-00-00-00



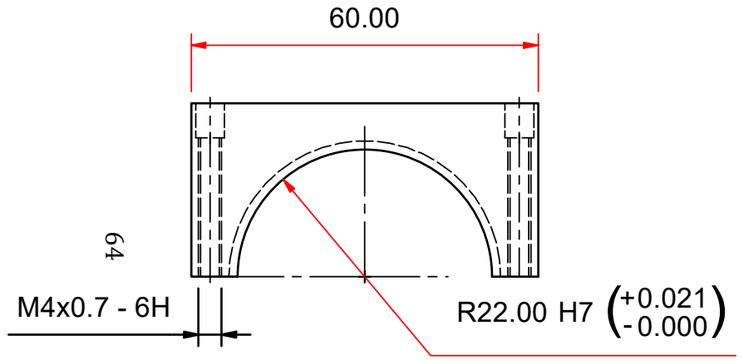
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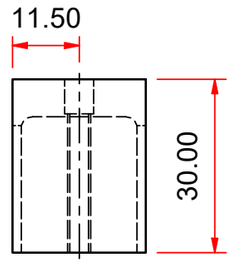
Parts List

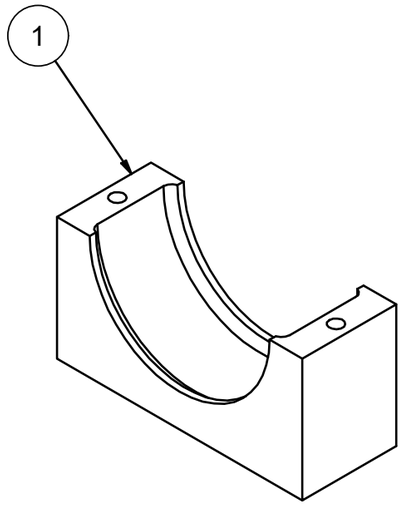
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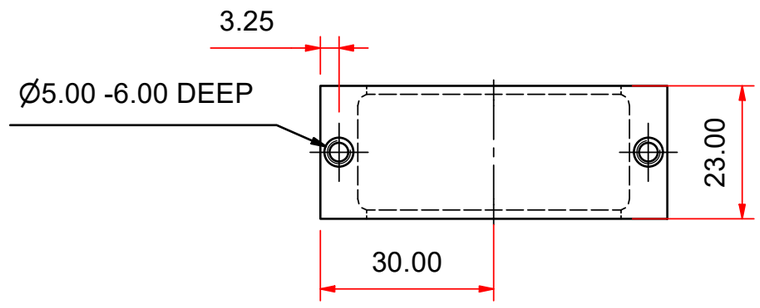
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STUDENT Nr. 24989142	**DRAWN BY** LN MATOTA	**CHECKED** K SCHREVE	**DATE** 2026/10/01	**SCALE ON A**3 = 1:2 **UNITS IN** mm	**TITLE:** JOINT SHAFT **SHEET Nr.** OF **SHEETS** **Nr.** 24989142-01-00-00







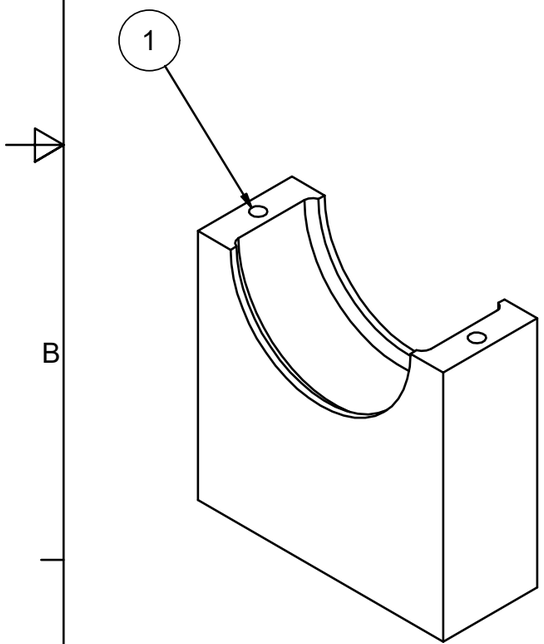


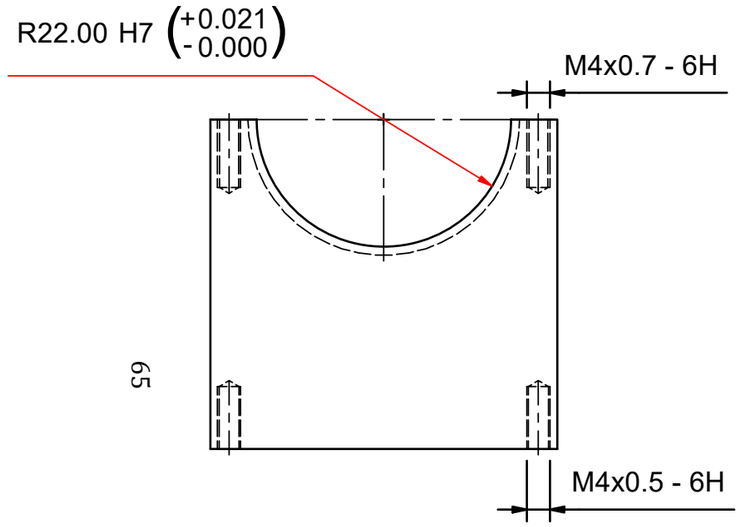
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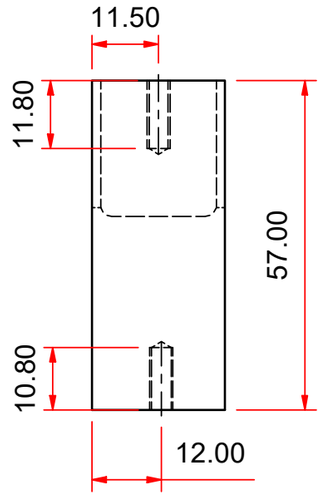
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ITEM	**DESCRIPTION**	**Qty.**	**MATERIAL / SPECIFICATIONS**

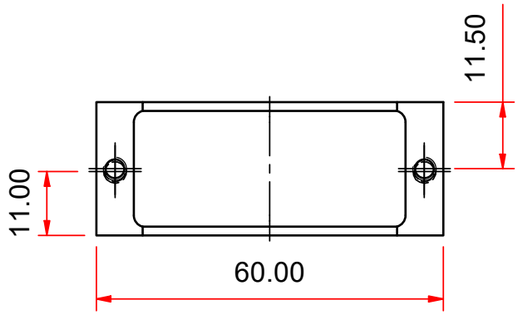
STELLENBOSCH UNIVERSITY

STUDENT Nr. 24989142	**DRAWN BY** LN MATOTA	**CHECKED** K SCHREVE	**SCALE ON A3** = 1:1 **UNITS IN** mm	**TITLE:** BEARING HOUSING TOP	**Nr.** 24989142-02-00-00
			DATE 2026/10/01	**SHEET Nr.** **OF** **SHEETS**	









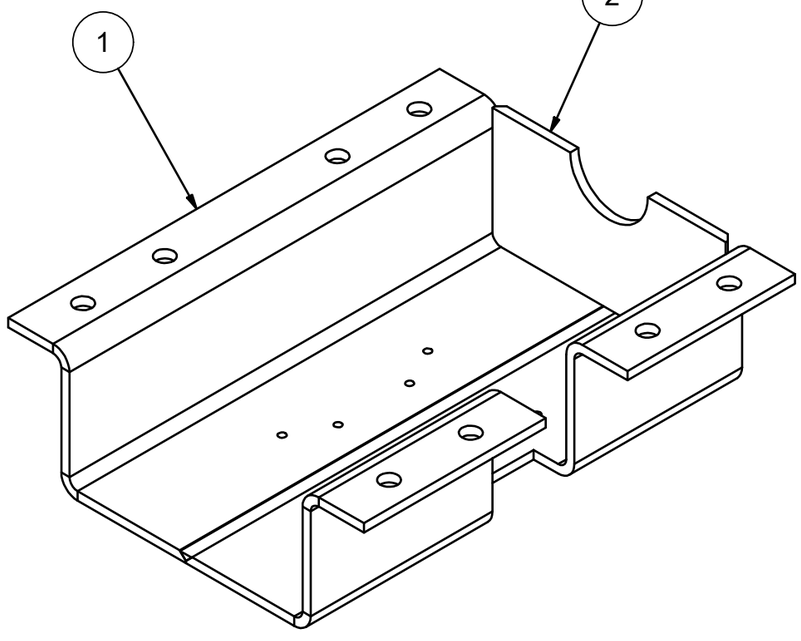
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ANGLES 1°

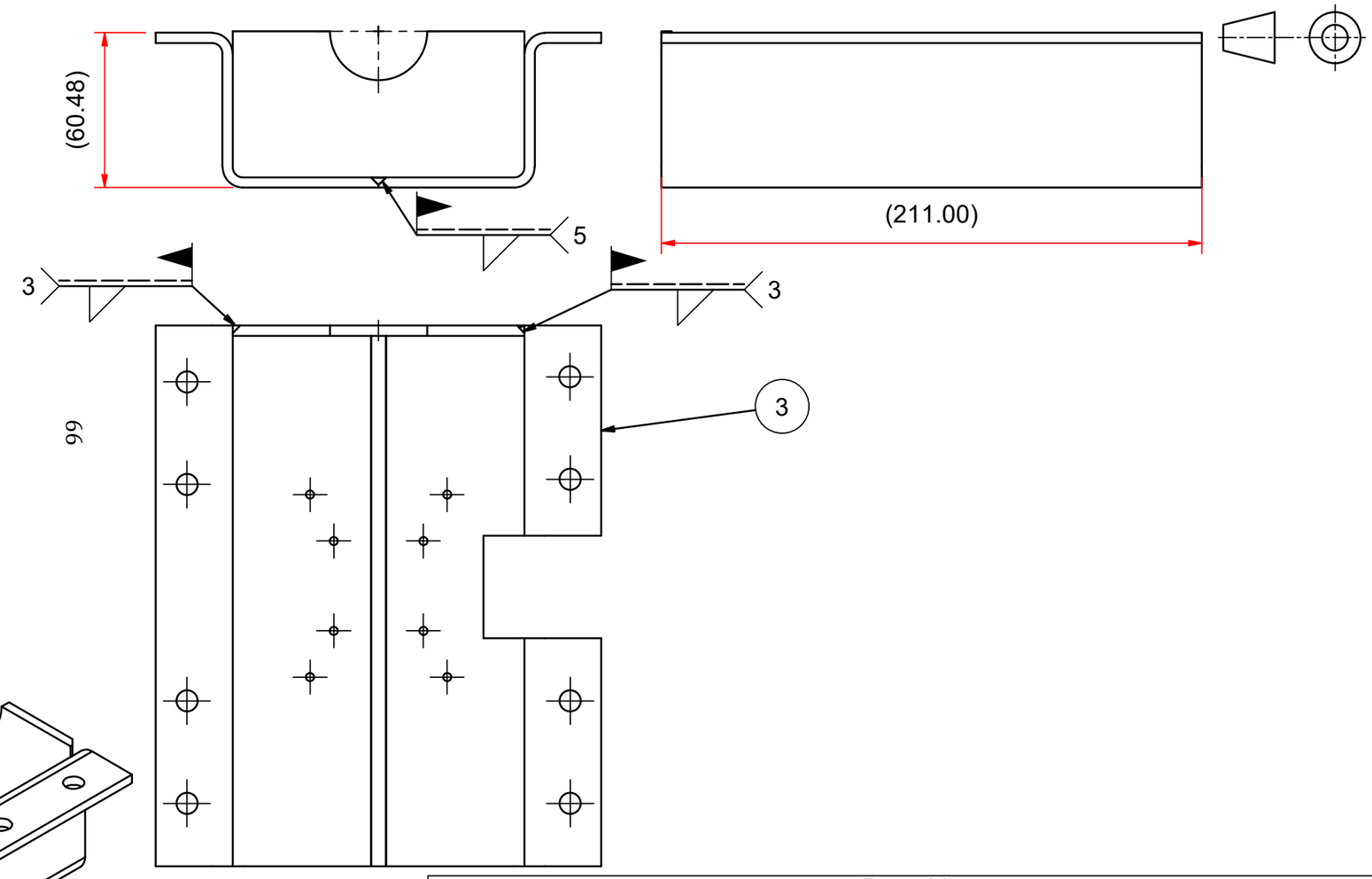
Parts List

1	24989142-03-00-00 BEARING HOUSING BOTTOM	2	ALUMINIUM
ITEM	**DESCRIPTION**	**Qty.**	**MATERIAL / SPECIFICATIONS**

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STUDENT Nr. 24989142	**DRAWN BY** LN MATOTA	**CHECKED** K SCHREVE	**DATE** 2026/10/01	**SCALE ON A3** = 1:1 **UNITS IN** mm	**TITLE:** BEARING HOUSING BOTTOM **SHEET Nr.** **OF** **SHEETS** **Nr.** 24989142-03-00-00





UNLESS OTHERWISE STATED
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ANGLES 1°

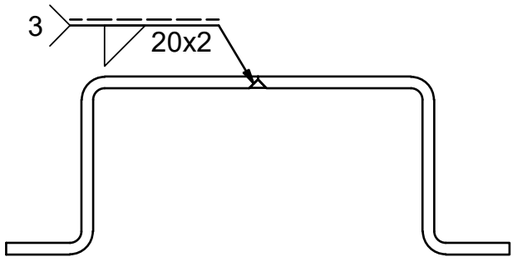
Parts List

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1	24989142-04-01-00 JOINT HOUSING BOTTON LEFT	1	ALUMINIUM

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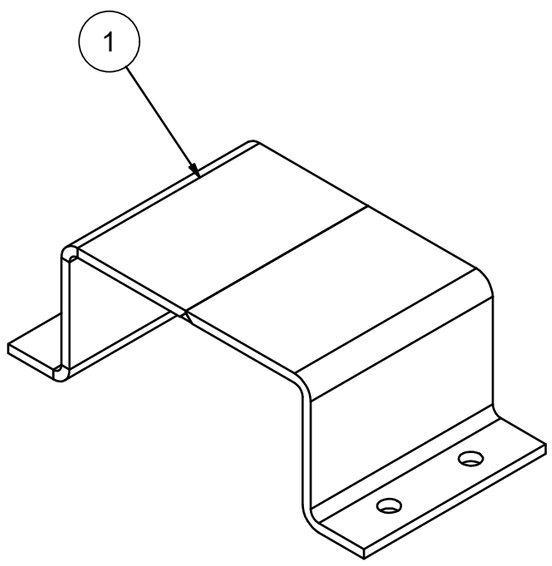
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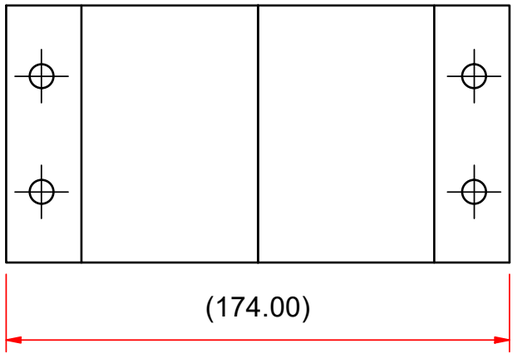
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67





UNLESS OTHERWISE STATED
TOLERANCES ± 0,1
ANGLES 1°

Parts List

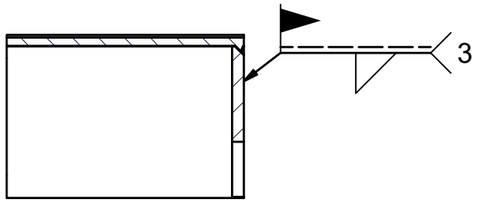
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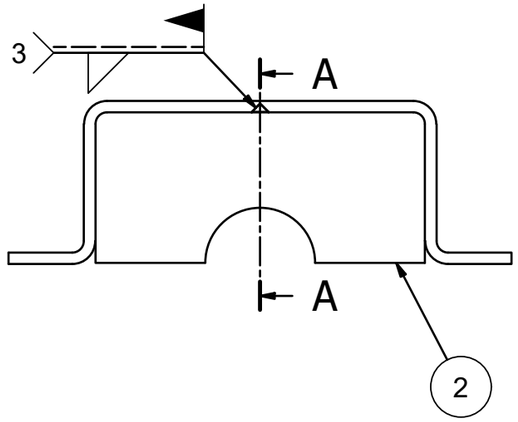
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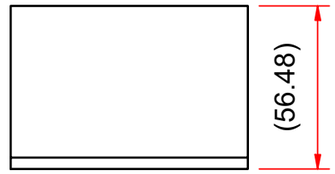
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UNITS IN mm
TITLE: JOINT HOUSING TOP-FRONT

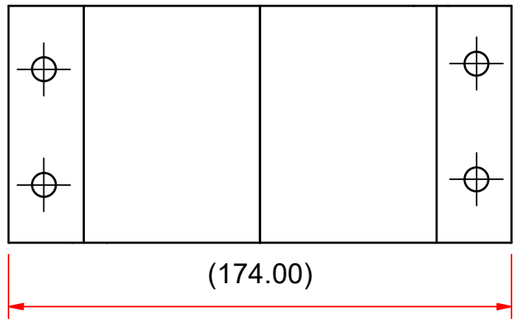
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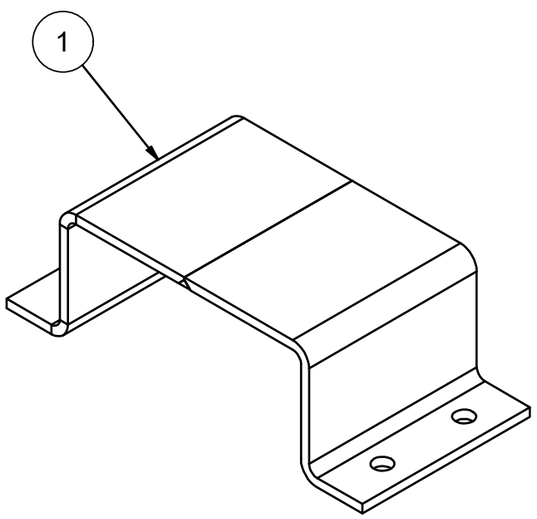
A-A (1 : 2)











89

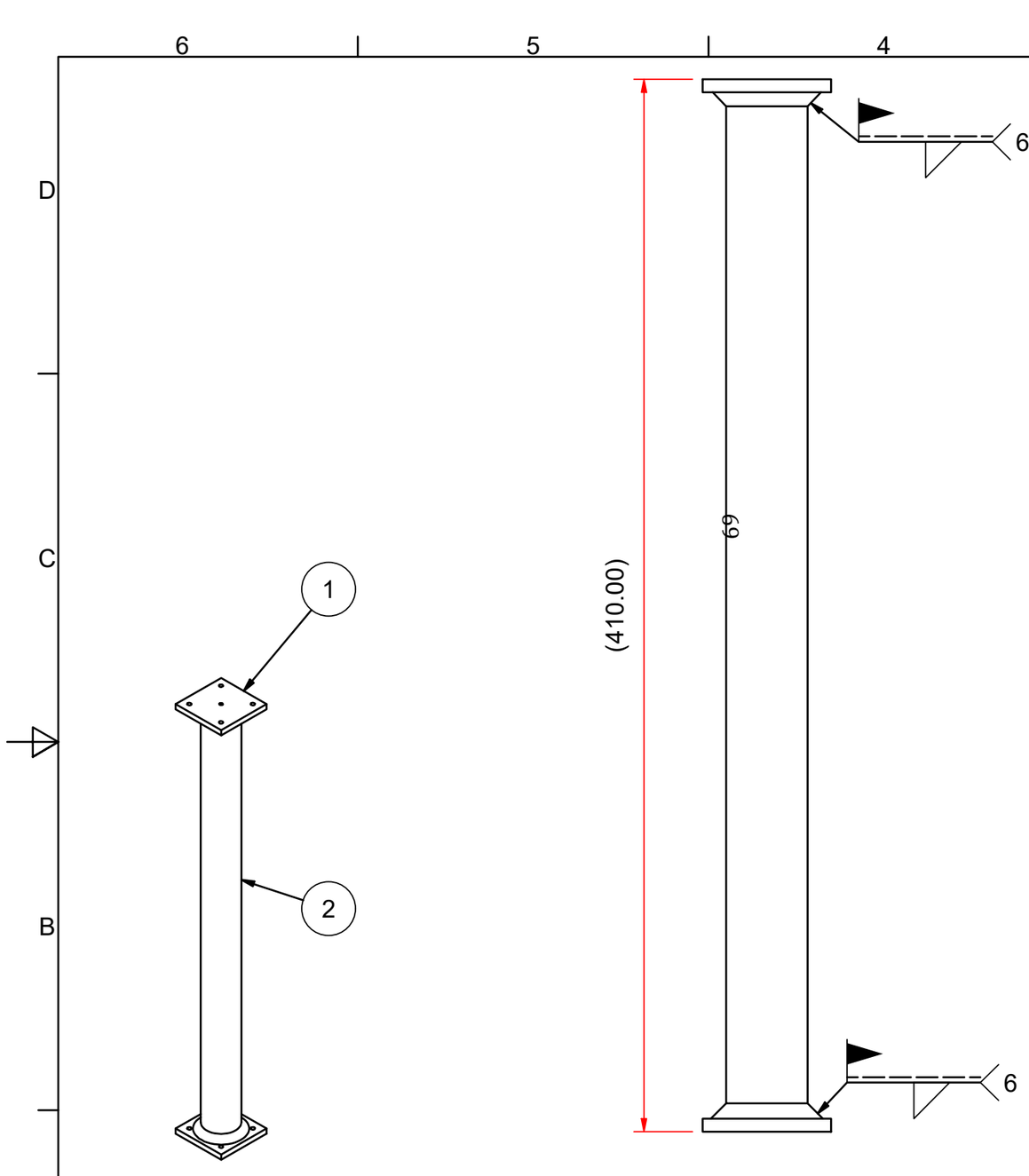
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ITEM	**DESCRIPTION**	**Qty.**	**MATERIAL / SPECIFICATIONS**

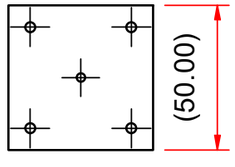
UNLESS OTHERWISE STATED
TOLERANCES ± 0,1
ANGLES 1˚

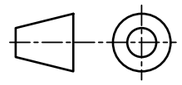
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SCALE ON A3 = 1:2
UNITS IN mm
TITLE: JOINT HOUSING TOP-BACK

STUDENT Nr. 24989142 | **DRAWN BY** LN MATOTA | **CHECKED** K SCHREVE | **DATE** 2026/10/01 | **SHEET Nr.** OF **SHEETS** | **Nr.** 24989142-06-00-00-00







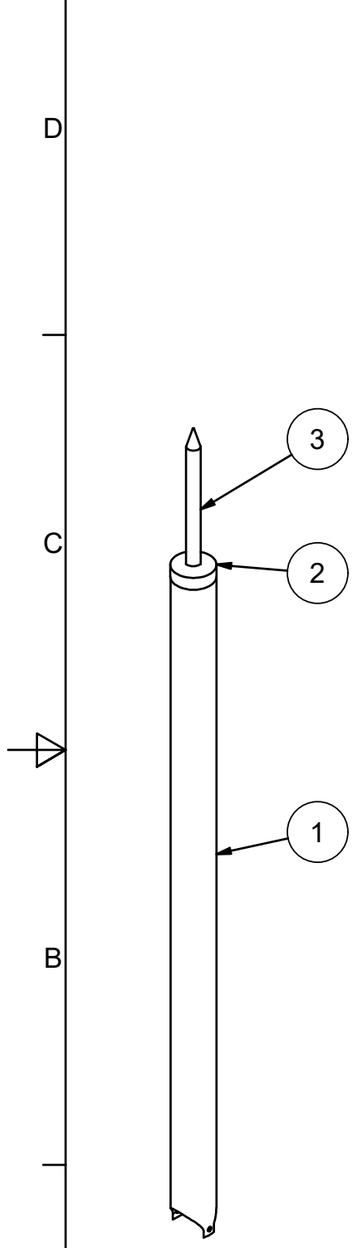
UNLESS OTHERWISE STATED
TOLERANCES ± 0,1
ANGLES 1°

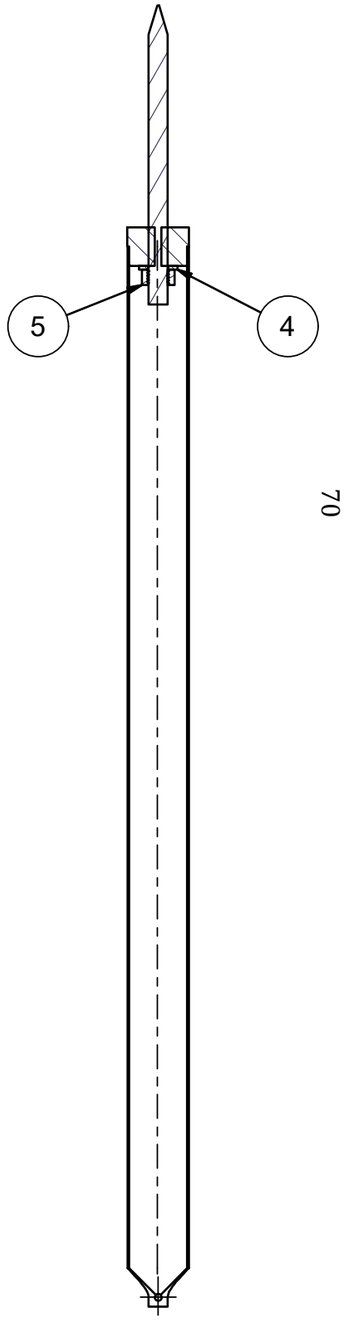
Parts List

2	24989142-07-02-00 BASE LINK PIPE	1	ALUMINIUM
1	24989142-07-01-00 BASE LINK PLATE	2	ALUMINIUM
ITEM	**DESCRIPTION**	**Qty.**	**MATERIAL / SPECIFICATIONS**

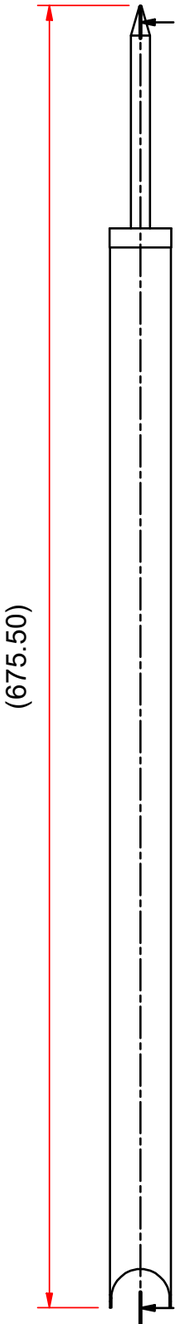
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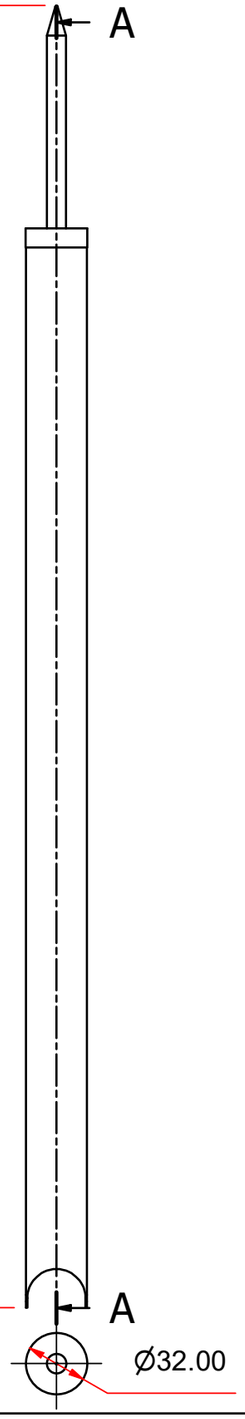
STUDENT Nr. 24989142	**DRAWN BY** LN MATOTA	**CHECKED** K SCHREVE	**DATE** 2026/10/01	**SCALE ON A3** = 1:2 **UNITS IN** mm	**TITLE:** BASE LINK **SHEET Nr.** **OF** **SHEETS** **Nr.** 24989142-07-00-00

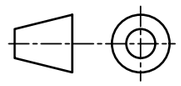




A-A (1 : 3)





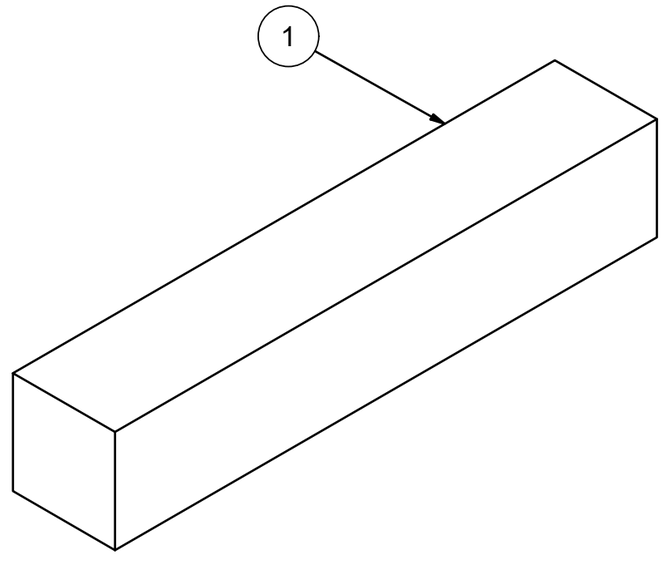


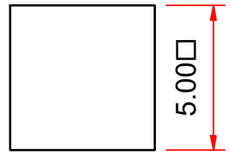
UNLESS OTHERWISE STATED
TOLERANCES ± 0,1
ANGLES 1°

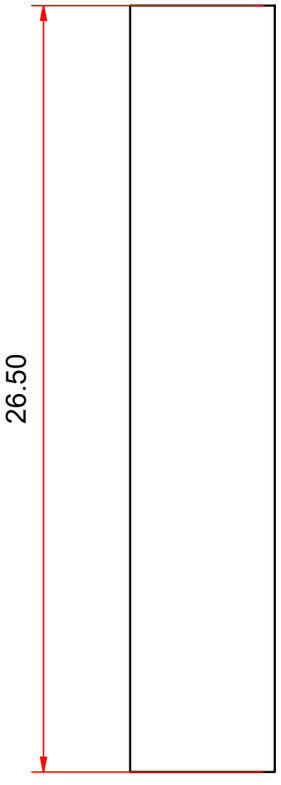
Parts List			
5	M10X1.5 NUT	1	-
4	WASHER ID 10MM	1	-
3	24989142-08-03-00 ARTICULATING LINK PROBE	1	ALUMINIUM
2	24989142-08-02-00 ARTICULATING LINK PROBE HUB	1	ALUMINIUM
1	24989142-08-01-00 ARTICULATING LINK PIPE	1	ALUMINIUM
ITEM	**DESCRIPTION**	**Qty.**	**MATERIAL / SPECIFICATIONS**

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STUDENT Nr. 24989142	**DRAWN BY** LN MATOTA	**CHECKED** K SCHREVE	**SCALE ON A3** = 1:3 **UNITS IN** mm	**TITLE:** ARTICULATING LINK	
			DATE 2026/10/01	**SHEET Nr.** OF **SHEETS**	**Nr.** 24989142-08-00-00







UNLESS OTHERWISE STATED
TOLERANCES ± 0,1
ANGLES 1°

Parts List

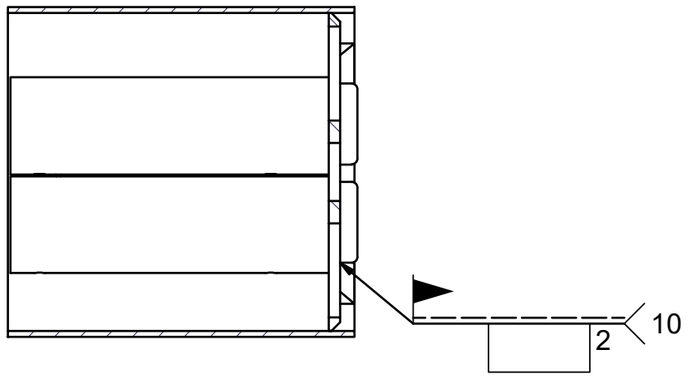
ITEM	DESCRIPTION	Qty.	MATERIAL / SPECIFICATIONS
1	24989142-09-00-00 BRAKE DISC KEY	1	BRASS

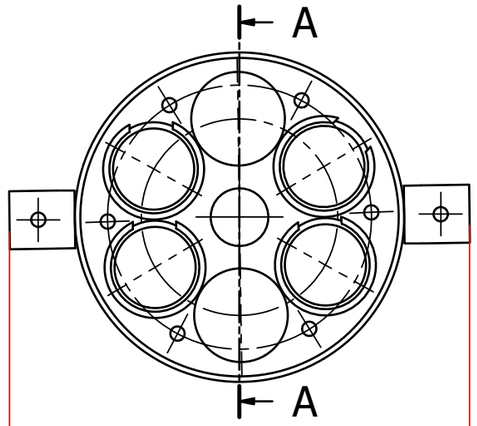
STELLENBOSCH UNIVERSITY

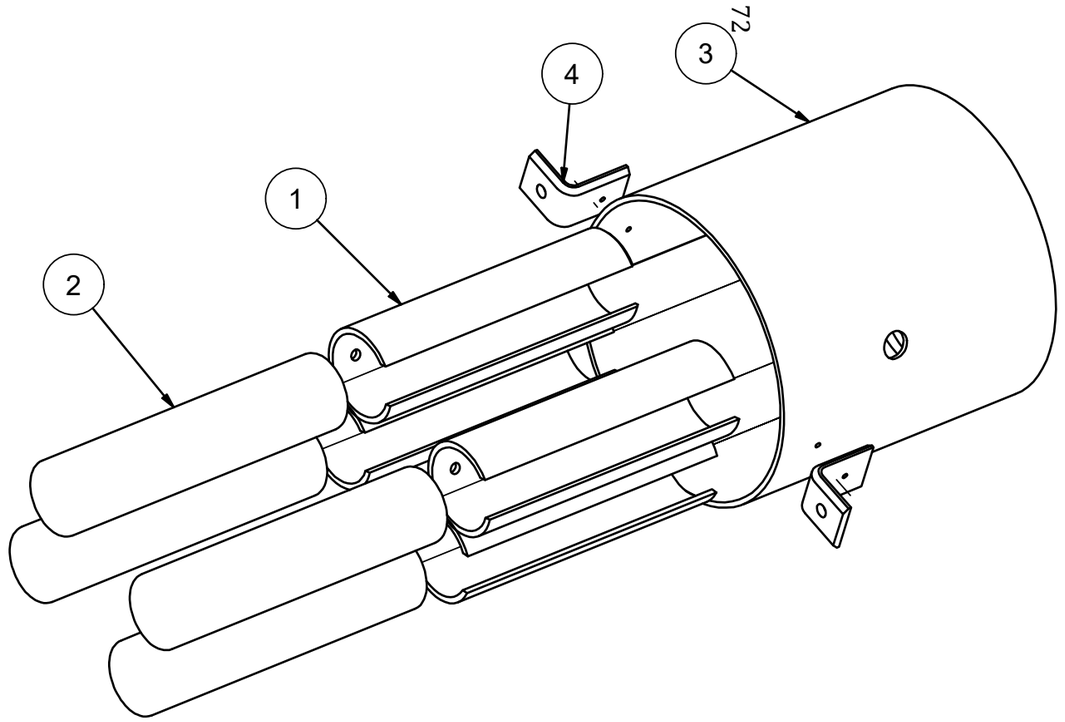
SCALE ON A3 = 5:1
UNITS IN mm
TITLE: BRAKE DISC KEY

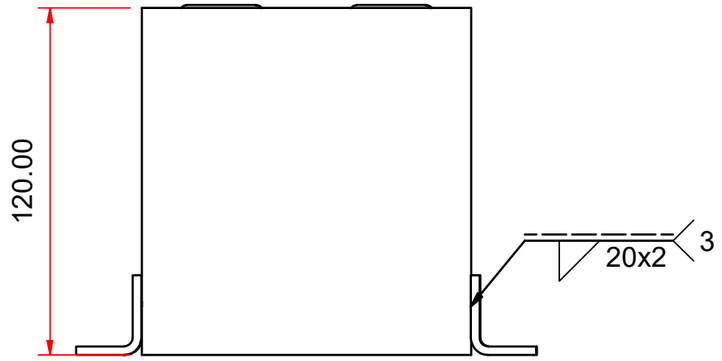
STUDENT Nr.	DRAWN BY	CHECKED	DATE	SHEET Nr. OF SHEETS	Nr.
24989142	LN MATOTA	K SCHREVE	2026/10/01		24989142-09-00-00

A-A (1 : 2)









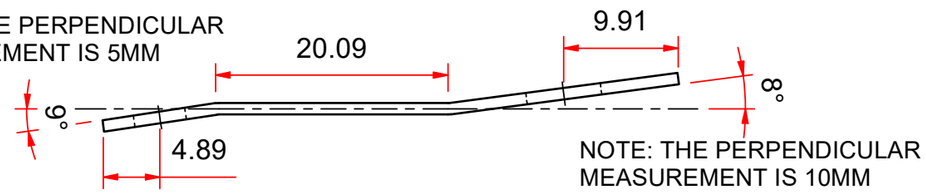
Parts List			
4	24989142-10-04-00 PRESSURE TABS	2	MILD STEEL
3	24989142-10-03-00 ELECTROMAGNETIC COVER	1	MILD STEEL
2	24989142-10-02-00 SOLENOID CORE	4	MILD STEEL
1	24989142-10-01-00 SOLENOID YOKE	4	MILD STEEL
ITEM	**DESCRIPTION**	**Qty.**	**MATERIAL / SPECIFICATIONS**

UNLESS OTHERWISE STATED
TOLERANCES ± 0,1
ANGLES 1°

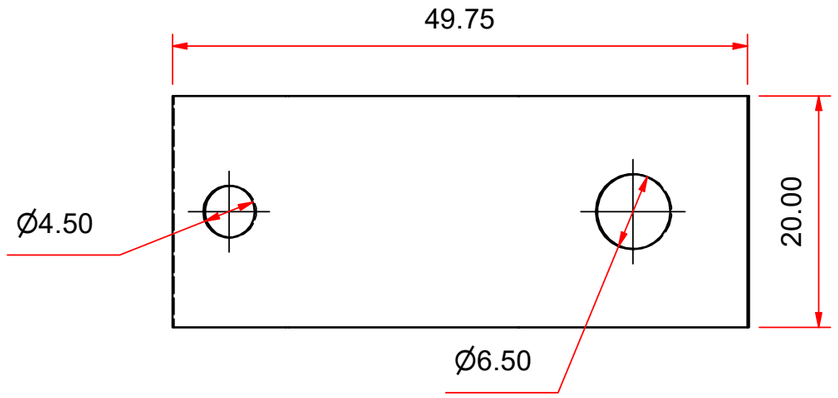
STELLENBOSCH UNIVERSITY

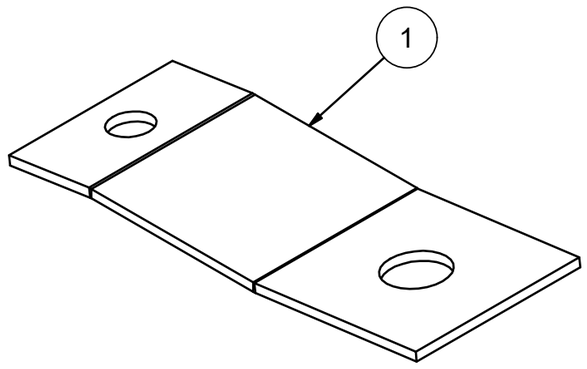
STUDENT Nr. 24989142	**DRAWN BY** LN MATOTA	**CHECKED** K SCHREVE	**DATE** 2026/10/01	**SCALE ON A3** = 1:2 **UNITS IN** mm	**TITLE:** ELECTROMAGNET **SHEET Nr.** **OF** **SHEETS** **Nr.** 24989142-10-00-00

NOTE: THE PERPENDICULAR
MEASUREMENT IS 5MM



NOTE: THE PERPENDICULAR
MEASUREMENT IS 10MM





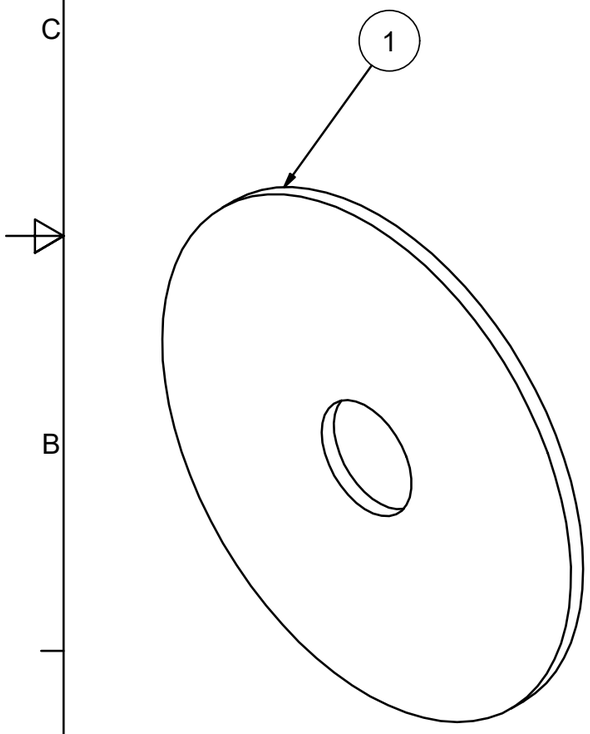
UNLESS OTHERWISE STATED
TOLERANCES ± 0,1
ANGLES 1°

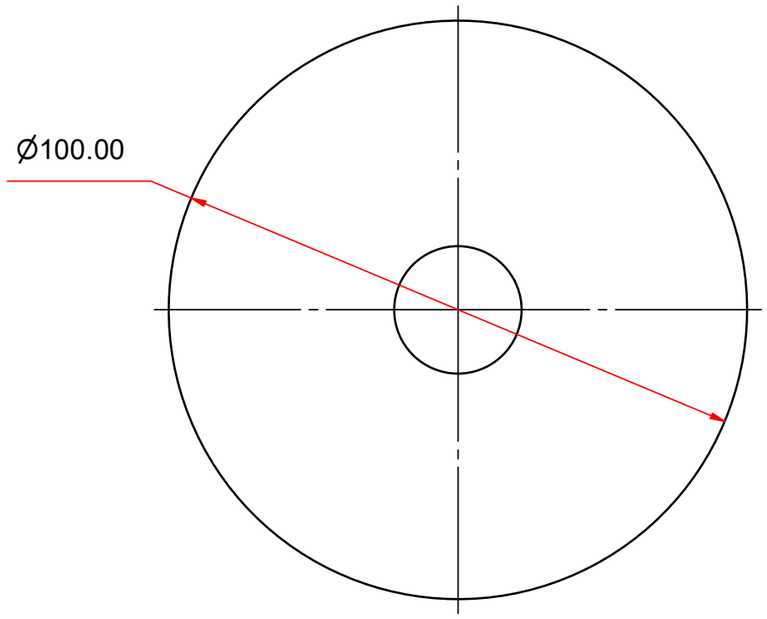
Parts List

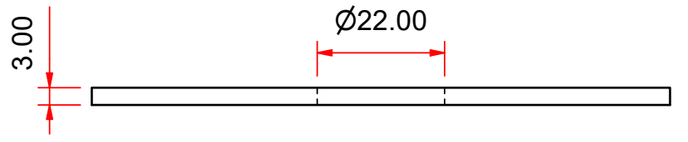
ITEM	DESCRIPTION	Qty.	MATERIAL / SPECIFICATIONS
1	24989142-11-00-00 PRESSURE TAB	1	MILD STEEL

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STUDENT Nr. 24989142	**DRAWN BY** LN MATOTA	**CHECKED** K SCHREVE	**SCALE ON A**3 = 2:1 **UNITS IN** mm	**TITLE:** PRESSURE TAB	**Nr.** 24989142-11-00-00
			DATE 2026/10/01	**SHEET Nr.** OF **SHEETS**	







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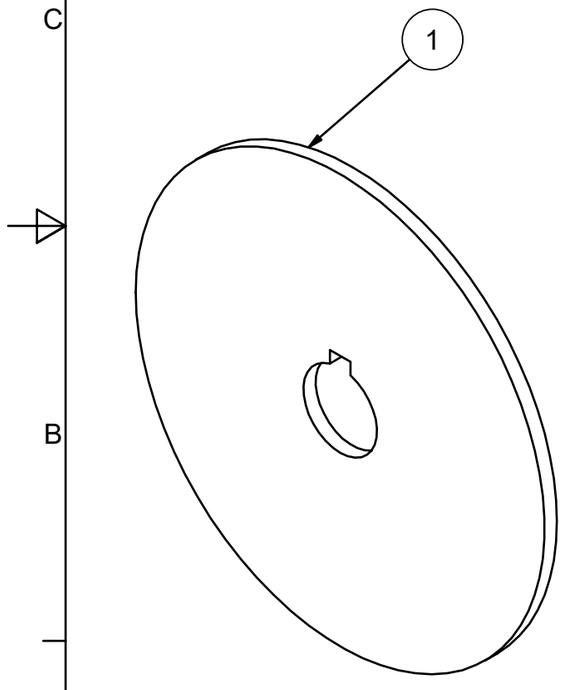
UNLESS OTHERWISE STATED
TOLERANCES ± 0,1
ANGLES 1°

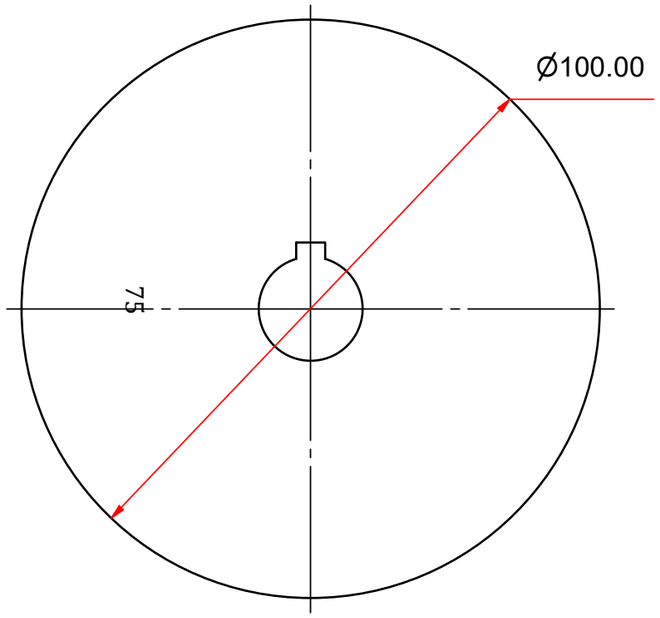
Parts List

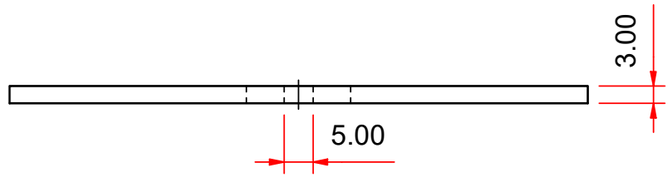
ITEM	DESCRIPTION	Qty.	MATERIAL / SPECIFICATIONS
1	24989142-14-00-00 FRICTION PLATE	1	MILD STEEL

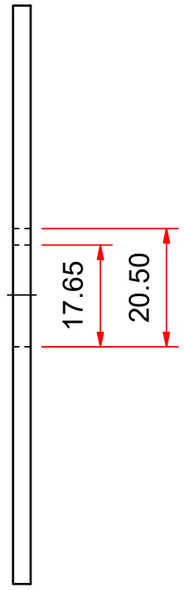
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STUDENT Nr. 24989142	**DRAWN BY** LN MATOTA	**CHECKED** K SCHREVE	**SCALE ON A**3 = 1:2 **UNITS IN** mm	**TITLE:** FRICTION PLATE	
			DATE 2026/10/01	**SHEET Nr.** OF **SHEETS**	**Nr.** 24989142-14-00-00









UNLESS OTHERWISE STATED
TOLERANCES ± 0,1
ANGLES 1°

Parts List

ITEM	DESCRIPTION	Qty.	MATERIAL / SPECIFICATIONS
1	24989142-12-00-00-00 BRAKE DISC	1	WOOD

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STUDENT Nr. 24989142	**DRAWN BY** LN MATOTA	**CHECKED** K SCHREVE	**SCALE ON A3** = 1:1 **UNITS IN** mm	**TITLE:** BRAKE DISC	**Nr.** 24989142-12-00-00-00
			DATE 2026/10/01	**SHEET Nr.** OF **SHEETS**	

Appendix D

Project Schedule

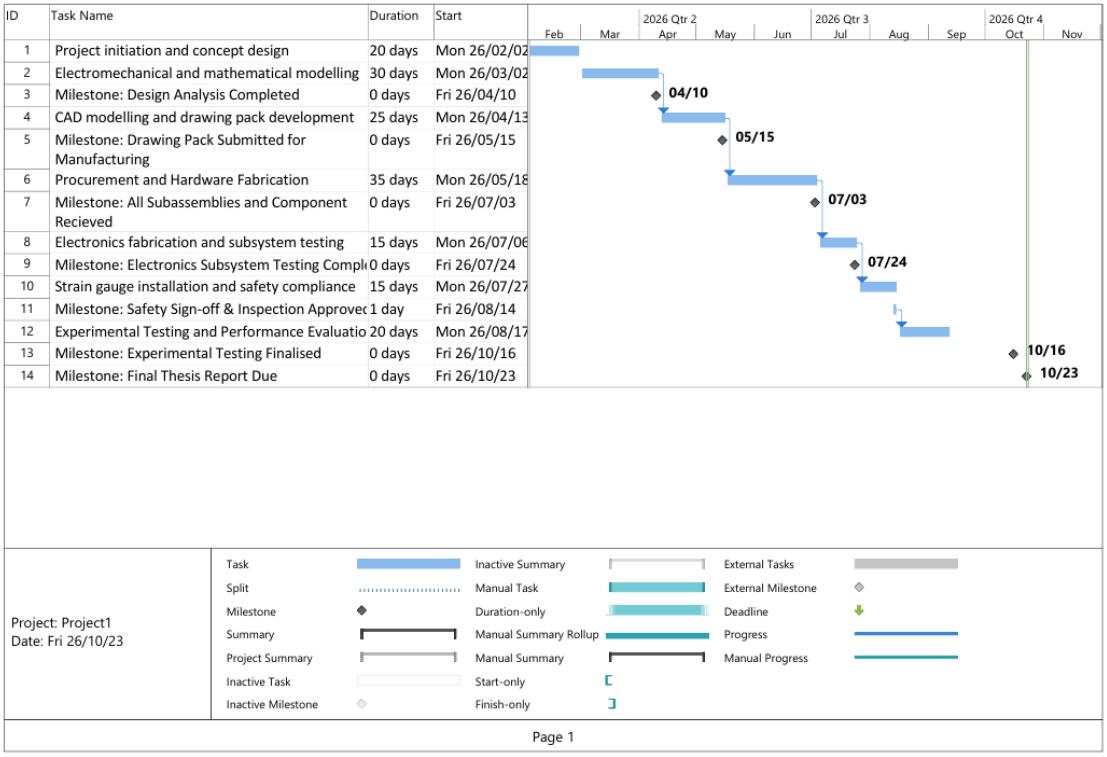

ID
Task Name
Duration
Start
1 Project initiation and concept design 20 days Mon 26/02/02
2 Electromechanical and mathematical modelling 30 days Mon 26/03/02
3 Milestone: Design Analysis Completed 0 days Fri 26/04/10
4 CAD modelling and drawing pack development 25 days Mon 26/04/13
5 Milestone: Drawing Pack Submitted for Manufacturing 0 days Fri 26/05/15
6 Procurement and Hardware Fabrication 35 days Mon 26/05/18
7 Milestone: All Subassemblies and Component Recieved 0 days Fri 26/07/03
8 Electronics fabrication and subsystem testing 15 days Mon 26/07/06
9 Milestone: Electronics Subsystem Testing Compl 0 days Fri 26/07/24
10 Strain gauge installation and safety compliance 15 days Mon 26/07/27
11 Milestone: Safety Sign-off & Inspection Approvec 1 day Fri 26/08/14
12 Experimental Testing and Performance Evaluatio 20 days Mon 26/08/17
13 Milestone: Experimental Testing Finalised 0 days Fri 26/10/16
14 Milestone: Final Thesis Report Due 0 days Fri 26/10/23
2026 Qtr 2
2026 Qtr 3
2026 Qtr 4
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
04/10
05/15
07/03
07/24
10/16
10/23
Project: Project1
Date: Fri 26/10/23
Task
Split
Milestone
Summary
Project Summary
Inactive Task
Inactive Milestone
Inactive Summary
Manual Task
Duration-only
Manual Summary Rollup
Manual Summary
Start-only
Finish-only
External Tasks
External Milestone
Deadline
Progress
Manual Progress
Page 1

Figure D.1

List of references

[1] L. Engineering, "The advantages of using a portable cmm arm in metrology," *Leussink Engineering*, 2024.

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[9] B. Alvarez, E. Cuesta, P. Luque, D. Mantaras, and D. Muina, "Dynamic deformations in coordinate measuring arms using virtual simulation," *Internation Journal of Simulation Modelling*, vol. 14, no. 4, pp. 609–620, 2015.

[10] *FaroArm Quantum X Series Series User Manual*, March 2024.

[11] HydraForce, Inc., *Solenoid Training Manual*. HydraForce, Inc., Lincolnshire, IL, USA, 1999.

[12] International Organisation for Standardisation, "Rolling bearings — radial bearings — geometrical product specifications (gps) and tolerance values," July 2014.

[13] R. C. Hibbler, *Mechanics of Materials*. Pearson, 10th ed., 2017.